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An Introduction to Martingales: Theory and Examples

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1. Introduction

The concept of martingale first originated as a betting strategy in the 18th century in France, which consisted of doubling the bet after every loss and leaving after the first win. In the 20th century, its name was given to a class of stochastic processes in probability theory that models a fair system, which has since influenced numerous areas of mathematics and applications. As Joseph L. Doob puts it, "Martingale theory illustrates the history of mathematical probability: the basic definitions are inspired by crude notions of gambling, but the theory has become a sophisticated tool of modern abstract mathematics, drawing from and contributing to other fields" [1].

This text provides an introduction to the theory of martingales, specifically focusing on martingales in discrete time. We will start by formalising the concept of conditional expectation, which provides the foundation for understanding how predictions change as new information is obtained over time. This concept provides the foundation for the definition of martingales, since the martingale property is expressed in terms of conditional expectations with respect to an evolving flow of information.

Next, we will introduce the concepts of martingale, submartingale and supermartingale, illustrating their properties with concrete examples and analysing which transformations preserve their structure. Following this, we will define the notions of predictable process and stopping times, leading us to explore key results and ultimately prove the upcrossing inequality and the martingale convergence theorem, which establishes the necessary conditions for almost sure convergence.

We will then explore stronger forms of convergence, firstly by discussing Doob's inequality, which naturally leads to convergence in $\mathcal{L}^p$, for $p > 1$. Afterwards, we turn to the case $p = 1$, which needs the notion of uniform integrability to give the additional control needed to obtain convergence in $\mathcal{L}^1$.

Moreover, we will examine optional stopping theorems. These results address one of the central questions in martingale theory: under what conditions can we stop martingales at random times while preserving their defining properties?

Finally, we apply the theory developed throughout the text to branching processes. In particular, we study the Galton–Watson process and show how martingales naturally arise when the population size is normalised by its expected growth. This provides a concrete example of how martingale convergence theorems can be used to analyse the long-term behaviour of a stochastic model.

This text is heavily inspired by Durrett's *Probability: Theory and Examples* [2]. Some elementary familiarity with measure theory is assumed from the reader, and the relevant measure-theoretic results used throughout the text are collected in the appendix.

2. Conditional Expectation

In order to formalise martingales, it is first necessary to introduce the concept of conditional expectation. In this section, we will define what a conditional expectation is and present some of its most important properties, as well as demonstrate its relevance with examples. Before giving the formal definition, we consider a simple example to motivate the concept.

Example 2.1 (Simple Random Walk) Let $\{X_n\}_{n\in\mathbb{N}_0}$ be a sequence of independent random variables such that $X_0 = 0$ and $X_n = \pm 1$ each with probability $1/2$. We can think of this as the movement of a particle on a horizontal line: the particle starts at position 0 and, at each step n, it either jumps one position to the left or one position to the right. Let $S_n = \sum_{i=0}^{n} X_i$ be the position of the particle after n steps.

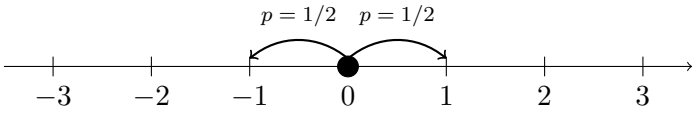


Figure 1: Simple random walk starting at $S_0 = 0$. At each step, the particle moves one position to the left or to the right with probability $1/2$.

We can easily conclude that $\mathbb{E}[X_n] = 0$, and using the linearity of expectation we can derive the expected position of the particle after n steps.

$$\mathbb{E}[S_n] = \mathbb{E}\left[\sum_{i=0}^{n} X_i\right] = \sum_{i=0}^{n} \mathbb{E}[X_i] = 0.$$

However, if we knew the position of the particle k steps after the initial position, how would that change our prediction of S_n for $n > k$? For example, suppose we know that $S_{n-1} = 100$. In this case, it is clear that $\mathbb{E}[S_n] = 0$ is an impossible prediction for S_n. Thus, the natural question that arises is: how can we formalise mathematically the idea of updating our prediction given the new information that we get? This motivates the formal definition of conditional expectation.

Definition 2.2 Let $(\Omega, \mathcal{F}_0, \mathbb{P})$ be a probability space, $\mathcal{F} \subseteq \mathcal{F}_0$ a σ-algebra, and $X \in \mathcal{F}_0$ a random variable with $\mathbb{E}[|X|] < \infty$. We define the conditional expectation of X given $\mathcal{F}$, denoted by $\mathbb{E}[X \mid \mathcal{F}]$, as any random variable Y such that:

1. $Y \in \mathcal{F}$, i.e. Y is $\mathcal{F}$-measurable;
2. $\int_A X \, \mathrm{d}\mathbb{P} = \int_A Y \, \mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{F}$.

Intuitively, we can think of $\mathcal{F}$ as the information we currently have, and $\mathbb{E}[X \mid \mathcal{F}]$ as the best prediction we can make given that information. In other words, conditional expectation is a mechanism to update our prediction of a random variable when new information becomes available. The following lemma ensures the integrability of a random variable satisfying the previous two conditions.

Lemma 2.3 *If Y satisfies both conditions from the definition of conditional expectation, then Y is integrable.*

Proof. For a proof, see [2], Section 4.1, Lemma 4.1.1. □

Although this lemma guarantees integrability, it does not yet show that such a random variable exists, nor that it is unique. The next result addresses these two points.

Proposition 2.4 *Let a random variable Y be the conditional expectation defined above. Then Y exists and is unique almost surely.*

Proof. For a proof, see Appendix A.1. □

This result means that different versions of Y may exist, however, any two such versions can differ only on sets with probability 0. Therefore, since these sets are negligible in probability, we can consider conditional expectation as a single object, up to almost sure equality.

Before exploring some examples, let us analyse some of the properties of conditional expectation.

Theorem 2.5 (Properties of Conditional Expectation) *Let X_n and X be integrable random variables, $\mathcal{F}, \mathcal{G} \subseteq \mathcal{F}_0$ σ-algebras, and $a \in \mathbb{R}$. Then:*

1. $\mathbb{E}[a\,X_1 + X_2 \mid \mathcal{F}] = a\,\mathbb{E}[X_1 \mid \mathcal{F}] + \mathbb{E}[X_2 \mid \mathcal{F}]$;
2. *If $X_1 \leq X_2$, then* $\mathbb{E}[X_1 \mid \mathcal{F}] \leq \mathbb{E}[X_2 \mid \mathcal{F}]$;
3. *If $X_n \geq 0$, $X_n \uparrow X$, and $\mathbb{E}[X] < \infty$, then* $\mathbb{E}[X_n \mid \mathcal{F}] \uparrow \mathbb{E}[X \mid \mathcal{F}]$;
4. *If $\mathcal{F} \subseteq \mathcal{G}$ and $\mathbb{E}[X \mid \mathcal{G}] \in \mathcal{F}$, then* $\mathbb{E}[X \mid \mathcal{F}] = \mathbb{E}[X \mid \mathcal{G}]$;
5. *If $\mathcal{F} \subseteq \mathcal{G}$, then*
   - *(i)* $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{F}]$;
   - *(ii)* $\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{F}] = \mathbb{E}[X \mid \mathcal{F}]$ *(Tower Property)*;
6. *If $X_1 \in \mathcal{F}$, then* $\mathbb{E}[X_1 X_2 \mid \mathcal{F}] = X_1\,\mathbb{E}[X_2 \mid \mathcal{F}]$;
7. $\mathbb{E}[X_1 X_2 \mid \mathcal{F}]^2 \leq \mathbb{E}[X_1^2 \mid \mathcal{F}]\,\mathbb{E}[X_2^2 \mid \mathcal{F}]$ *(Cauchy-Schwarz inequality).*
8. *If X is independent of $\mathcal{F}$, then* $\mathbb{E}[X \mid \mathcal{F}] = \mathbb{E}[X]$

Proof. For a proof for 1 to 4, see [2], Section 4.1.2, Theorems 4.1.9 and 4.1.12. For a proof for 5 to 8, see Appendix A.2. □

Properties 5(ii), 6 and 8 are particularly interesting. Conceptually, property 5(ii), or most commonly known as the tower property, implies that if a prediction of X is first conditioned on the larger set, and then conditioned on a smaller set, it yields the prediction of X using only the smaller set. From an intuitive point of view, if we consider $\mathcal{F}$ to be the information available today and $\mathcal{G}$ the information available tomorrow, then the tower property says that, predicting today what tomorrow's best prediction will be, gives the same result as the best prediction from today's information. This result will later be extremely useful to extend the martingale property.

Moreover, properties 6 and 8 have a very intuitive interpretation. Since $X_1 \in \mathcal{F}$, that is, X_1 is known given the information in $\mathcal{F}$, then X_1 behaves like a deterministic quantity relative to this conditioning, hence, it can be taken out of the conditional expectation. On the other hand, if X is independent of $\mathcal{F}$, it means that $\mathcal{F}$ does not contain any additional information about X, thus conditioning on $\mathcal{F}$ does not change our prediction of X.

Example 2.6 Let us recall the random walk from Example 2.1. We left the example wondering how knowing S_{n-1} would change our prediction of S_n, that is, what is the value of $\mathbb{E}[S_n \mid S_{n-1}]$? After some simple calculations, we can see that

$$\mathbb{E}[S_n \mid S_{n-1}] = \mathbb{E}[S_{n-1} + X_n \mid S_{n-1}] = \mathbb{E}[S_{n-1} \mid S_{n-1}] + \mathbb{E}[X_n \mid S_{n-1}].$$

Here, S_{n-1} is already known because the expectation is conditioned on S_{n-1}, and together with property 8 from Theorem 2.5 we have

$$\mathbb{E}[S_n \mid S_{n-1}] = S_{n-1} + \mathbb{E}[X_n] = S_{n-1}.$$

As we can see, in this case, knowing more information changes the prediction from the unconditional expectation $\mathbb{E}[S_n] = 0$ to the conditional expectation S_{n-1}. In fact, we have just concluded that the best prediction of the next position, given the information currently available, is the current position itself, which is exactly the defining property of a martingale.

3. Martingales, Submartingales and Supermartingales

Now we are ready to formalise the concept of a martingale. In this section, we will define the concepts of martingale, submartingale and supermartingale, as well as explore some of their key properties. Moreover, we will introduce the notions of predictable processes and stopping times, which will be crucial to construct and analyse different types of martingales. These will be essential to prove the upcrossing inequality, which allows us to control the fluctuations of martingales. Finally, we will present the martingale convergence theorem, which provides the necessary conditions for almost sure convergence.

3.1. Definition and some transformations

As previously mentioned, a martingale describes a fair process, more specifically, it is a stochastic process whose best prediction for the next value, given all the information available at present, is the current value. Consequently, they are essential in the study of random processes that have no systematic tendencies. Naturally, martingales are built upon the notion of conditional expectation. However, in order to condition on the information available at each time, we first need a way of representing how information evolves over time.

Definition 3.1 A collection of σ-algebras $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$ is a filtration on $(\Omega, \mathcal{F}, \mathbb{P})$ if

$$\mathcal{F}_m \subseteq \mathcal{F}_n \subseteq \mathcal{F}, \quad \forall m \leq n,$$

i.e. it is an increasing sequence of σ-algebras. We say that $\mathcal{F}_m$ is coarser than $\mathcal{F}_n$ and that $\mathcal{F}_n$ is finer than $\mathcal{F}_m$.

A filtration should be understood as the information available over time. The σ-algebra $\mathcal{F}_n$ represents the information available up to time n. Since later times contain at least as much information as earlier times, it is natural that the sequence of σ-algebras is increasing. This allows us to formalise the idea of making predictions based only on the information available at the present time, which is precisely the framework needed to define martingales.

Definition 3.2 Let $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$ be a filtration. The stochastic process $(X_n)_{n\in\mathbb{N}_0}$ is a martingale (in discrete time) with respect to the filtration $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$ if it satisfies the following conditions:

1. $\mathbb{E}[|X_n|] < \infty$, for all n;
2. X_n is adapted to $\mathcal{F}_n$, i.e. $X_n \in \mathcal{F}_n$ for all n;
3. $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$, for all n (Martingale Property).

Proposition 3.3 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a martingale with respect to the filtration $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$. Then, for all n we have*

$$\mathbb{E}[X_n] = \mathbb{E}[X_0].$$

Proof. For a proof, see Appendix A.3. □

The martingale property ensures that the process is fair, meaning that there are no systematic trends. However, this condition can be relaxed to describe processes with positive or negative tendencies (drift), which are called submartingale and supermartingale, respectively.

Definition 3.4 Let $(X_n)_{n\in\mathbb{N}_0}$ be an adapted integrable process with respect to $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$. Then

- $(X_n)_{n\in\mathbb{N}_0}$ is a submartingale if $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n$ for all n;
- $(X_n)_{n\in\mathbb{N}_0}$ is a supermartingale if $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n$ for all n.

Note: Throughout this text, all martingales, submartingales and supermartingales are considered with respect to the filtration $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$, unless stated otherwise.

Proposition 3.5 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a submartingale. Then, for all n we have*

$$\mathbb{E}[X_n] \geq \mathbb{E}[X_0].$$

Similarly, if $(X_n)_{n\in\mathbb{N}_0}$ is a supermartingale, then for all n we have

$$\mathbb{E}[X_n] \leq \mathbb{E}[X_0].$$

Proof. The proof is analogous to the proof for Proposition 3.3, only with $\geq$ and $\leq$ instead of $=$, respectively. □

Submartingales and supermartingales generalise the concept of a martingale to allow for drift, which can be useful when dealing with biased processes.

In reality, the martingale property can be generalised to any future time. In other words, for any submartingale or supermartingale, the conditional expectation of a future value, given the present information, is bounded above or below by the current value. The following theorem formalises this extension.

Theorem 3.6 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a submartingale. Then, for all $n > m$,*

$$\mathbb{E}[X_n \mid \mathcal{F}_m] \geq X_m.$$

Similarly, if $(X_n)_{n\in\mathbb{N}_0}$ is a supermartingale, then

$$\mathbb{E}[X_n \mid \mathcal{F}_m] \leq X_m.$$

The martingale case follows immediately as the special case where equality holds:

$$\mathbb{E}[X_n \mid \mathcal{F}_m] = X_m.$$

Proof. For a proof, see [2], Section 4.2, Theorems 4.2.4 and 4.2.5. There is also a simple, yet elegant alternative proof for the martingale case that I would like to show. For $n > m$, we have that $\mathcal{F}_m \subseteq \mathcal{F}_{n-1}$, thus by the tower property we have

$$\mathbb{E}[X_n \mid \mathcal{F}_m] = \mathbb{E}[\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] \mid \mathcal{F}_m] = \mathbb{E}[X_{n-1} \mid \mathcal{F}_m].$$

We can repeat this step until we get

$$\mathbb{E}[X_n \mid \mathcal{F}_m] = \mathbb{E}[X_{n-1} \mid \mathcal{F}_m] = \cdots = \mathbb{E}[X_{m+1} \mid \mathcal{F}_m] = X_m,$$

which ends the proof. □

Example 3.7 Let us recall Example 2.1 and show that $(S_n)_{n\in\mathbb{N}_0}$ is a martingale. Firstly, we can easily see that $|S_n| \leq n$, and so

$$\mathbb{E}[|S_n|] \leq n < \infty.$$

Moreover, let us consider $(\mathcal{F}_n)_{n\in\mathbb{N}_0} = \sigma(X_1, \ldots, X_n)$. Then $S_n = \sum_{i=0}^{n} X_i \in \mathcal{F}_n$, because $X_1, \ldots, X_n \in \mathcal{F}_n$. Hence, S_n is adapted to $\mathcal{F}_n$. Lastly,

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[S_n + X_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[S_n \mid \mathcal{F}_n] + \mathbb{E}[X_{n+1}] = S_n.$$

Therefore, $(S_n)_{n\in\mathbb{N}_0}$ is a martingale.

What if we were to change the probabilities of moving left or right at each step? How would this affect the expected position and the martingale property of the process? Let us consider that $X_n = 1$ with probability p, and $X_n = -1$ with probability $1 - p$, with $p \neq 1/2$. Then, the expected value of X_n would be

$$\mathbb{E}[X_n] = p - (1 - p) = 2p - 1.$$

Hence, we have

$$\mathbb{E}[S_{n+1} \mid \mathcal{F}_n] = \mathbb{E}[S_n \mid \mathcal{F}_n] + \mathbb{E}[X_{n+1}] = S_n + 2p - 1.$$

Therefore, $(S_n)_{n\in\mathbb{N}_0}$ is a submartingale if $p > 1/2$, meaning it has a positive drift, thus the particle is expected to move further to the right over time. On the other hand, if $p < 1/2$, then $(S_n)_{n\in\mathbb{N}_0}$ is a supermartingale, that is, it has negative drift and is expected to move further to the left over time.

We now turn our attention to transformations of martingales and analyse which transformations preserve or modify the martingale property.

Lemma 3.8 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a martingale and $a, b \in \mathbb{R}$. Then $(aX_n + b)_{n\in\mathbb{N}_0}$ is also a martingale.*

Proof. Since $X_n \in \mathcal{F}_n$, then $aX_n + b \in \mathcal{F}_n$. Moreover,

$$\mathbb{E}[|aX_n + b|] \leq |a|\, \mathbb{E}[|X_n|] + |b| < \infty.$$

Finally, by linearity of conditional expectation,

$$\mathbb{E}[aX_{n+1} + b \mid \mathcal{F}_n] = a\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] + b = aX_n + b.$$

which ends the proof. □

As the previous lemma shows, linear transformations preserve the martingale property. A natural follow-up question is whether nonlinear transformations can also preserve or partially preserve it. In particular, as we will see in the next theorem, convex functions generate submartingales.

Theorem 3.9 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a martingale and φ a convex function with $\mathbb{E}[|\varphi(X_n)|] < \infty$ for all n. Then $(\varphi(X_n))_{n\in\mathbb{N}_0}$ is a submartingale.*

Proof. For a proof, see [2], Section 4.2., Theorem 4.2.6. □

This theorem can be extended to submartingales with increasing convex functions and supermartingales with increasing concave functions.

Theorem 3.10 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a submartingale and φ an increasing convex function with $\mathbb{E}[|\varphi(X_n)|] < \infty$ for all n. Then $(\varphi(X_n))_{n\in\mathbb{N}_0}$ is a submartingale.*

Proof. For a proof, see [2], Section 4.2., Theorem 4.2.7. □

Corollary 3.11 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a supermartingale and φ an increasing concave function with $\mathbb{E}[|\varphi(X_n)|] < \infty$ for all n. Then $(\varphi(X_n))_{n\in\mathbb{N}_0}$ is a supermartingale.*

Proof. For a proof, see Appendix A.4. □

Consequently, from Theorem 3.9, it directly follows that if $\mathbb{E}[|X_n|^p] < \infty$, then $(|X_n|^p)_{n\in\mathbb{N}_0}$ is a submartingale for all $p \geq 1$. Similarly, from Theorem 3.10, it follows that if $(X_n)_{n\in\mathbb{N}_0}$ is a submartingale, then $\left((X_n - a)^+\right)_{n\in\mathbb{N}_0}$ is a submartingale. Moreover, by Corollary 3.11, if $(X_n)_{n\in\mathbb{N}_0}$ is a supermartingale, then $(X_n \wedge a)_{n\in\mathbb{N}_0}$ is a supermartingale. These particular cases will later be useful to prove the upcrossing inequality and the martingale convergence theorem.

3.2. Predictable processes and stopping times

To further study new types of martingales, we will need to introduce two new concepts. Firstly, we will want to describe processes whose values at each time step can be determined solely from past information, without looking into the future, thereby motivating the notion of predictable processes.

Definition 3.12 A stochastic process $(H_n)_{n\in\mathbb{N}_0}$ is predictable if $H_n \in \mathcal{F}_{n-1}$, for all $n \geq 1$.

Proposition 3.13 *Let $(H_n)_{n\in\mathbb{N}_0}$ be predictable and integrable. Then,*

$$\mathbb{E}[H_n \mid \mathcal{F}_{n-1}] = H_n.$$

Proof. It follows directly from property 6 of Theorem 2.5. □

In other words, H_n is determined before observing the new randomness at time n. For instance, we can think of it as the amount of money one will bet on a game of roulette at time n. The amount will need to be decided before the next round starts, that is, H_n is $\mathcal{F}_{n-1}$-measurable.

Next, consider the case where the gambler wants to stop once a certain threshold is reached. How can we formalise this notion? More generally, this represents any rule that decides when to stop based only on the information up to the current time, which is exactly the idea of a stopping time.

Definition 3.14 A random variable N is a stopping time if $\{N \leq n\} \in \mathcal{F}_n$, for all $n < \infty$.

Put simply, N is a stopping time only if we can decide whether it has already occurred, allowing us to observe a process at random times without using future information. Naturally, one should ask whether the structure is preserved after stopping a process.

Theorem 3.15 *Let N be a stopping time and let $(X_n)_{n\in\mathbb{N}_0}$ be a martingale. Then $(X_{N\wedge n})_{n\in\mathbb{N}_0}$ is a martingale. The same result holds for submartingales and supermartingales.*

Proof. For a proof, see Appendix A.5. □

This result is fundamental because it states that stopping a martingale preserves its structure, allowing us to analyse martingales at random times.

Example 3.16 (The Martingale Strategy) Let us consider the process $(Y_n)_{n\in\mathbb{N}_0}$, such that

$$Y_n = \sum_{i=1}^{n} H_i X_i,$$

with $Y_0 = 0$, and where $(X_n)_{n\in\mathbb{N}_0}$ is the random walk from Example 2.1 and

$$H_n = \begin{cases} 2^{n-1} & \text{if } X_1 = \cdots = X_{n-1} = -1, \\ 0 & \text{otherwise.} \end{cases}$$

We can think of X_n as the outcome of the nth round of a game, where $X_n = 1$ represents a win and $X_n = -1$ represents a loss, and H_n as the amount we bet at step n. This system is exactly the strategy known in the gambling world as a martingale, where we double the amount we bet after each loss and drop out of the game immediately after a win. We can see that, if the player has not won after n rounds,

i.e. if $X_1 = \cdots = X_n = -1$, the total loss is

$$\sum_{i=1}^{n} 2^{i-1} = 2^n - 1.$$

Therefore, if the player wins the $(n+1)^{\text{th}}$ round, we have

$$Y_{n+1} = Y_n + H_{n+1} = -(2^n - 1) + 2^n = 1,$$

meaning that stopping immediately after winning compensates all previous losses and leaves the player with a net profit of one unit.

Let us now verify whether this strategy aligns with its mathematical counterpart. Firstly, we check integrability. We can see that

$$\mathbb{E}[|H_n X_n|] = 2^{n-1} \cdot \left(\frac{1}{2}\right)^{n-1} = 2^{n-1-n+1} = 1,$$

and therefore for each n

$$\mathbb{E}[|Y_n|] \leq \mathbb{E}\left[\sum_{i=1}^{n} |H_i X_i|\right] = \sum_{i=1}^{n} \mathbb{E}[|H_i X_i|] = n < \infty,$$

thus, Y_n is integrable.

To verify that Y_n is adapted, we need to realise that $(H_n)_{n\in\mathbb{N}_0}$ is predictable, since H_n only depends on $X_1, \ldots, X_{n-1}$, which are all $\mathcal{F}_{n-1}$-measurable. Intuitively, we will only bet on round n if we lost in round $n-1$, meaning that the decision to bet is solely based on the information available up until round $n-1$. Therefore, since $\mathcal{F}_{n-1} \subseteq \mathcal{F}_n$, H_n is also $\mathcal{F}_n$-measurable. Moreover, $X_n \in \mathcal{F}_n$, hence $H_n X_n$ is $\mathcal{F}_n$-measurable and, since $\mathcal{F}_m \subseteq \mathcal{F}_n$ for all $m < n$, we have that $Y_n \in \mathcal{F}_n$.

Lastly, let us verify that it satisfies the martingale property.

$$\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] = \mathbb{E}\left[\sum_{i=1}^{n+1} H_i X_i \,\middle|\, \mathcal{F}_n\right] = \sum_{i=1}^{n} \mathbb{E}[H_i X_i \mid \mathcal{F}_n] + \mathbb{E}[H_{n+1} X_{n+1} \mid \mathcal{F}_n].$$

We know that $H_i X_i \in \mathcal{F}_n$ for all $i \leq n$, thus

$$\sum_{i=1}^{n} \mathbb{E}[H_i X_i \mid \mathcal{F}_n] = \sum_{i=1}^{n} H_i X_i = Y_n.$$

For this process to satisfy the martingale property, we just need to conclude that $\mathbb{E}[H_{n+1} X_{n+1} \mid \mathcal{F}_n] = 0$. For that, let us recall once again that $(H_n)_{n\in\mathbb{N}_0}$ is predictable and that X_{n+1} is independent from all past realisations. Hence, it follows that

$$\mathbb{E}[H_{n+1} X_{n+1} \mid \mathcal{F}_n] = H_{n+1} \cdot \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = H_{n+1} \cdot \mathbb{E}[X_{n+1}] = 0.$$

Example 3.17 Let us consider again the martingale strategy from the previous example, and let $N = \inf\{n \geq 1 : X_n = 1\}$, that is, N is the first time the player wins a round. We can easily verify that N is a stopping time, as for each $n \geq 1$

$$\{N \leq n\} = \cup_{i=1}^{n} \{Y_i = 1\} \in \mathcal{F}_n.$$

This shows that a stopping time depends only on the information available up to the present time. The player does not need to know the future in order to decide whether to continue or stop: at round n, they only need to know whether a win has occurred by that time.

By Proposition 3.3, for every $n \geq 0$,

$$\mathbb{E}[Y_n] = \mathbb{E}[Y_0] = 0,$$

and we may expect that the same conclusion would remain valid if the instant n is replaced by a random instant N. However, as we will see, this is not true in this example.

The probability that the first win occurs exactly at time n is

$$\mathbb{P}\{N = n\} = \left(\frac{1}{2}\right)^n,$$

thus,

$$\mathbb{P}\{N < \infty\} = \sum_{n=1}^{\infty} \mathbb{P}\{N = n\} = \sum_{n=1}^{\infty} \left(\frac{1}{2}\right)^n = 1.$$

Therefore, $N < \infty$ almost surely, that is, with probability one, the player eventually stops after a finite number of rounds. Moreover, by construction of the strategy, at N the player has recovered all previous losses and has gained one unit, so

$$Y_N = 1 \quad \text{a.s.}$$

and consequently

$$\mathbb{E}[Y_N] = 1.$$

This seemingly suggests that this strategy guarantees the player a profit of one unit. However, N is not bounded. In fact, for all $m \geq 1$,

$$\mathbb{P}\{N > m\} = \left(\frac{1}{2}\right)^m > 0.$$

Intuitively, this means that, although the game ends in finite time, this strategy requires the possibility of playing for an arbitrarily long time. Similarly, on each realised path where a win occurs, only a finite amount of money is needed, yet there is no finite bound on how much might be needed. Therefore, this strategy is only possible under idealised conditions, but is physically unrealistic.

In section 5, we will explore under what conditions n can be replaced by a stopping time N and still conclude that $\mathbb{E}[Y_N] = \mathbb{E}[Y_0]$.

3.3. Upcrossing inequality and martingale convergence theorem

We will now construct the necessary setup to introduce and prove the upcrossing inequality. Suppose that $(X_n)_{n\in\mathbb{N}_0}$ is a submartingale, $a < b \in \mathbb{R}$ and $N_0 = -1$. For $k \geq 0$, let

$$N_{2k-1} = \inf\{m > N_{2k-2} : X_m \leq a\} \quad \text{and} \quad N_{2k} = \inf\{m > N_{2k-1} : X_m \geq b\}.$$

N_{2k-1} is the first time after N_{2k-2} that the process reaches or goes below a, while N_{2k} is the first time after that at which the process reaches or goes above b. The intervals $(N_{2k-1}, N_{2k}]$ represent the periods during which the process moves from level a to level b, which are called upcrossings. Moreover, the N_j are stopping times, since the events $\{N_j \leq n\}$ depend only on the values of the process up to time n.

We now define the following strategy:

$$H_n = \begin{cases} 1 & \text{if } N_{2k-1} < n \leq N_{2k} \text{ for some } k, \\ 0 & \text{otherwise,} \end{cases}$$

which is a predictable process, since $\{N_{2k-1} < n \leq N_{2k}\} = \{N_{2k-1} \leq n-1\} \cap \{N_{2k} \leq n-1\}^c \in \mathcal{F}_{n-1}$. Intuitively, we can think of this strategy as buying a stock when its price is below a and selling it once it surpasses b, thus making a profit of at least $(b-a)$ per upcrossing. Additionally, let us define

$$U_n = \sup\{k : N_{2k} \leq n\},$$

which represents the number of completed upcrossings of $[a, b]$ by time n.

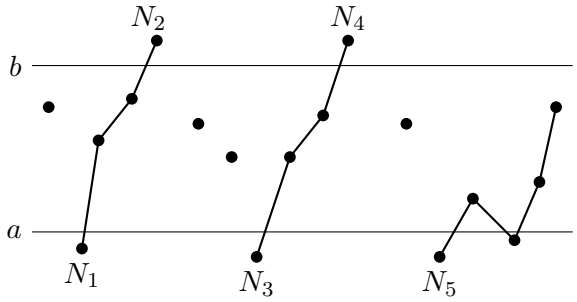


Figure 2: Upcrossings of the interval $[a, b]$.

Lastly, note that if one bets according to a gambling system H, then their winnings at time n can be modelled by

$$(H \cdot X)_n = \sum_{i=1}^{n} H_i(X_i - X_{i-1}),$$

where X_n is the net amount of money one would have won at time n if one had bet one dollar each time. In this example, since $H_n \in \{0, 1\}$, then H_n is nonnegative and bounded. The following theorem shows that such strategies preserve the supermartingale property, which will be the main tool for proving the upcrossing inequality.

Theorem 3.18 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a supermartingale. If $H_n \geq 0$ is predictable and each H_n is bounded, then $((H \cdot X)_n)_{n\in\mathbb{N}_0}$ is a supermartingale.*

Proof. For a proof, see [2], Section 4.2, Theorem 4.2.8. □

Now, we are ready to state and prove the upcrossing inequality.

Theorem 3.19 (Upcrossing Inequality) *Let $(X_m)_{m\in\mathbb{N}_0}$ be a submartingale and let $a, b \in \mathbb{R}$ with $a < b$. Denote by U_n the number of upcrossings of the interval $[a, b]$ completed by the process up to time n. Then*

$$(b-a)\,\mathbb{E}[U_n] \leq \mathbb{E}[(X_n - a)^+] - \mathbb{E}[(X_0 - a)^+].$$

Proof. Let $Y_m = a + (X_m - a)^+$. By Theorem 3.10, it follows that Y_m is also a submartingale. This transformation of X_m pushes all values below a up to exactly a and therefore it does not change the number of upcrossings, that is, Y_m and X_m have the same number of upcrossings of the interval $[a, b]$. Each upcrossing results in a profit of at least $(b - a)$. If there is an incomplete final upcrossing, let us show that it still makes a non-negative contribution to $(H \cdot Y)_n$. Since the upcrossing is incomplete, then we have that $n \in (N_{2k-1}, N_{2k})$, for some k. In the interval $(N_{2k-1}, n]$, we hold one stock, meaning that its contribution to $(H \cdot Y)_n$ is

$$Y_n - Y_{N_{2k-1}} = Y_n - a \geq 0.$$

Hence, it follows that

$$(b-a)\,U_n \leq (H \cdot Y)_n \implies (b-a)\,\mathbb{E}[U_n] \leq \mathbb{E}[(H \cdot Y)_n].$$

Now, let $K_m = 1 - H_m$. Then

$$(H \cdot Y)_n + (K \cdot Y)_n = \sum_{i=1}^{n} H_i(Y_i - Y_{i-1}) + \sum_{i=1}^{n}(1 - H_i)(Y_i - Y_{i-1}) = \sum_{i=1}^{n}(Y_i - Y_{i-1}) = Y_n - Y_0.$$

Since Y_m is a submartingale, then $-Y_m$ is a supermartingale. Moreover, K is non-negative, predictable and bounded. Therefore, by Theorem 3.18, we have that $(K \cdot (-Y))_n$ is a supermartingale. However,

$$(K \cdot (-Y))_n = -\sum_{i=1}^{n} K_i(Y_i - Y_{i-1}) = -(K \cdot Y)_n$$

Hence, $(K \cdot Y)_n$ is a submartingale and it follows that

$$\mathbb{E}[(K \cdot Y)_n] \geq \mathbb{E}[(K \cdot Y)_0] = 0.$$

Therefore,

$$\mathbb{E}[(H \cdot Y)_n] \leq \mathbb{E}[Y_n - Y_0] = \mathbb{E}[(X_n - a)^+] - \mathbb{E}[(X_0 - a)^+],$$

proving the inequality. □

The upcrossing inequality gives an upper bound on the expected number of times a submartingale can cross from below a to above b. More specifically,

$$\mathbb{E}[U_n] \leq \frac{\mathbb{E}[(X_n - a)^+] - \mathbb{E}[(X_0 - a)^+]}{(b-a)}.$$

Therefore, if the expected positive part remains bounded, then the process cannot cross repeatedly between two fixed levels forever, which leads naturally to the next result.

Theorem 3.20 (Martingale Convergence Theorem) *Let $(X_n)_{n\in\mathbb{N}_0}$ be a submartingale such that $\sup_n \mathbb{E}[X_n^+] < \infty$. Then $\lim_n X_n = X$ almost surely, with $\mathbb{E}[|X|] < \infty$.*

Proof. First, let us note that

$$(X_n - a)^+ \leq X_n^+ + |a| \implies \mathbb{E}[(X_n - a)^+] \leq \mathbb{E}[X_n^+] + |a|,$$

and that

$$\mathbb{E}[(X_0 - a)^+] \geq 0.$$

Therefore, by Theorem 3.19, we have

$$\mathbb{E}[U_n] \leq \frac{\mathbb{E}[(X_n - a)^+] - \mathbb{E}[(X_0 - a)^+]}{(b-a)} \leq \frac{\mathbb{E}[X_n^+] + |a|}{(b-a)},$$

and, since $\sup_n \mathbb{E}[X_n^+] < \infty$, it follows that $\sup_n \mathbb{E}[U_n] < \infty$. Now, let U be the total number of upcrossings of the interval $[a, b]$. Clearly, $U_n \geq 0$ and $U_n \uparrow U$, hence, by the Monotone Convergence Theorem (see [3], section 2.2, Theorem 2.14), we have

$$\mathbb{E}[U] = \lim_n \mathbb{E}[U_n] < \infty.$$

Thus, $U < \infty$ a.s., i.e. $\mathbb{P}\{U = \infty\} = 0$, which means that there is only a finite number of upcrossings made by the process. Define

$$A_{a,b} := \left\{ \liminf_n X_n < a < b < \limsup_n X_n \right\}.$$

On $A_{a,b}$, the process upcrosses the interval $[a, b]$ infinitely many times. Thus, $A_{a,b} \subseteq \{U = \infty\}$ and, consequently, $\mathbb{P}\{A_{a,b}\} = 0$. Now, consider de set $D = \{(a, b) \in \mathbb{Q}^2 : a < b\}$. Since this set is countable, we have

$$\mathbb{P}\left\{\cup_{(a,b)\in D} A_{a,b}\right\} \leq \sum_{(a,b)\in D} \mathbb{P}\{A_{a,b}\} = 0.$$

Hence, $\mathbb{P}\left\{\cup_{(a,b)\in D} A_{a,b}\right\} = 0$. To prove that the process converges, we will need to show that $\lim_n X_n$ exists a.s., i.e. $\liminf_n X_n = \limsup_n X_n$ a.s. Suppose that $\liminf_n X_n < \limsup_n X_n$. By the density of

$\mathbb{Q}$ in $\mathbb{R}$, there exist $(a, b) \in D$ such that

$$\liminf_n X_n < a < b < \limsup_n X_n.$$

Hence,

$$\left\{\liminf_n X_n < \limsup_n X_n\right\} \subseteq \bigcup_{(a,b)\in D} \left\{\liminf_n X_n < a < b < \limsup_n X_n\right\} = \bigcup_{(a,b)\in D} A_{a,b}$$

Since $\mathbb{P}\left\{\cup_{(a,b)\in D} A_{a,b}\right\} = 0$, we have that

$$\mathbb{P}\left\{\liminf_n X_n < \limsup_n X_n\right\} = 0,$$

and, therefore, $\liminf_n X_n = \limsup_n X_n$ a.s.

Now we prove that X is integrable. Note that we can write $X = X^+ - X^-$, where $X^+ = (X)^+$ and $X^- = (-X)^+$. Since $X_n \to X$ a.s., then $X_n^+ \to X^+$ a.s., hence, by Fatou's lemma (see [3], Section 2.2, Theorem 2.18),

$$\mathbb{E}[X^+] \leq \liminf_n \mathbb{E}[X_n^+] < \infty,$$

thus, $X < \infty$ a.s. On the other hand, $X_n^- \to X^-$ a.s. and note that

$$\mathbb{E}[X_n^-] = \mathbb{E}[X_n^+] - \mathbb{E}[X_n] \leq \mathbb{E}[X_n^+] - \mathbb{E}[X_0],$$

where the last inequality results from the fact that $(X_n)_{n\in\mathbb{N}_0}$ is a submartingale. Once again, we can apply Fatou's lemma, and it follows that

$$\mathbb{E}[X^-] \leq \liminf_n \mathbb{E}[X_n^-] \leq \sup_n \mathbb{E}[X_n^+] - \mathbb{E}[X_0] < \infty.$$

Hence, $\mathbb{E}[|X|] < \infty$. □

This result shows that a simple boundedness condition is enough to guarantee almost sure convergence of a submartingale, and that it converges to a well-defined random variable. The key idea comes from the upcrossing inequality: under this boundedness condition, the process cannot oscillate between two fixed levels infinitely many times. Therefore, it must eventually settle down and converge. A particularly important case is obtained when the process is a non-negative supermartingale.

Corollary 3.21 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a non-negative supermartingale. Then*

$$\lim_n X_n = X \ \text{ a.s. and } \mathbb{E}[X] \leq \mathbb{E}[X_0].$$

Proof. For a proof, see [2], Section 4.2, Theorem 4.2.12. □

This corollary shows that non-negativity of a supermartingale is sufficient to guarantee almost sure

convergence. Nevertheless, both results fail to guarantee stronger forms of convergence, as we will see in the following example.

Example 3.22 Let S_n be the simple random walk from Example 2.1, but with initial position 1, i.e. $S_0 = 1$. Let $N = \inf\{n : S_n = 0\}$, that is, the first time the particle reaches position 0, and $Y_n = S_{N \wedge n}$. Since $(S_n)_{n \in \mathbb{N}_0}$ is a martingale and N a stopping time, by Theorem 3.15, we know that $(Y_n)_{n \in \mathbb{N}_0}$ is also a martingale. In fact, it is also non-negative, since the walk stops once it first reaches 0. Therefore, from Corollary 3.21 it follows that Y_n converges to a random variable $Y < \infty$ almost surely. We can clearly conclude that convergence to any $k > 0$ is not possible, since if $Y_n = k$ then $Y_{n+1} = k \pm 1$, thus $Y = 0$ almost surely. However, $(Y_n)_{n \in \mathbb{N}_0}$ is a martingale, so by Proposition 3.3 we have

$$\mathbb{E}[Y_n] = \mathbb{E}[Y_0] = 1.$$

Therefore, it follows that for all n

$$\int_\Omega |Y_n - Y| \, \mathrm{d}\mathbb{P} = \int_\Omega Y_n \, \mathrm{d}\mathbb{P} = \mathbb{E}[Y_n] = 1.$$

Hence,

$$\lim_n \int_\Omega |Y_n - Y| \, \mathrm{d}\mathbb{P} = 1 \neq 0$$

i.e. $Y_n \not\to Y$ in $\mathcal{L}^1$. This example illustrates that, although the martingale convergence theorem guarantees almost sure convergence, it may not be sufficient to guarantee convergence in $\mathcal{L}^p$. In the next section, we will explore which conditions guarantee convergence in $\mathcal{L}^p$.

4. Integrability Conditions for Martingale Convergence

In the previous chapter, we presented the necessary conditions for almost sure convergence with the martingale convergence theorem. In this section, we will go one step further and explore convergence in $\mathcal{L}^p$. We begin with Doob's inequality, which provides control over the size of martingales, and eventually leads to the $\mathcal{L}^p$ convergence theorem, which establishes the necessary conditions for a martingale to converge in $\mathcal{L}^p$, for $p > 1$. Then, we will introduce the concept of uniform integrability, which is the key condition for convergence in $\mathcal{L}^1$ to occur.

4.1. Doob's inequality and convergence in $\mathcal{L}^p$, for $p > 1$

Before proving Doob's inequality, we present a simple bounded version of an optional stopping theorem. Although more general optional stopping results will be discussed later, this finite-time version is enough for the arguments in this section.

Theorem 4.1 *Let $(X_n)_{n\in\mathbb{N}}$ be a submartingale and let N be a stopping time with $\mathbb{P}\{N \le k\} = 1$. Then*

$$\mathbb{E}[X_0] \le \mathbb{E}[X_N] \le \mathbb{E}[X_k].$$

Proof. For a proof, see [2], Section 4.4, Theorem 4.4.1. □

In words, this result guarantees that the submartingale property is preserved when stopping a process at a bounded stopping time. This bounded form of optional stopping will be used in the proof of Doob's inequality.

Theorem 4.2 (Doob's Inequality) *Let $(X_n)_{n\in\mathbb{N}}$ be a submartingale and define $\bar{X}_n = \max_{0\le m\le n} X_m^+$. Let $A = \{\bar{X}_n \ge \lambda\}$ for $\lambda > 0$. Then*

$$\lambda\,\mathbb{P}\{A\} \le \mathbb{E}[X_n \mathbb{1}_A] \le \mathbb{E}[X_n^+].$$

Proof. We start by proving the first inequality. Let $N = \inf\{m : X_m \ge \lambda \text{ or } m = n\}$. Since $X_N \ge \lambda$ on A, then

$$\lambda\mathbb{P}\{A\} \le \mathbb{E}[X_N \mathbb{1}_A].$$

Note that

$$\mathbb{E}[X_N] = \mathbb{E}[X_N \mathbb{1}_A] + \mathbb{E}[X_N \mathbb{1}_{A^c}] \quad \text{and} \quad \mathbb{E}[X_n] = \mathbb{E}[X_n \mathbb{1}_A] + \mathbb{E}[X_n \mathbb{1}_{A^c}].$$

Since $X_N = X_n$ on A^c, we have

$$\mathbb{E}[X_N \mathbb{1}_{A^c}] = \mathbb{E}[X_n \mathbb{1}_{A^c}].$$

Therefore, from Theorem 4.1, it follows that

$$\mathbb{E}[X_N] \leq \mathbb{E}[X_n] \Leftrightarrow \mathbb{E}[X_N \mathbb{1}_A] \leq \mathbb{E}[X_n \mathbb{1}_A].$$

The second inequality follows from the fact that $X_n \mathbb{1}_A \leq X_n^+$. Hence, $\mathbb{E}[X_n \mathbb{1}_A] \leq \mathbb{E}[X_n^+]$. □

Doob's inequality provides an upper bound for the probability that the process crosses a certain level λ before time n. Intuitively, it answers the question: how likely is it that the process becomes larger than λ before time n? And the answer to that question is

$$\mathbb{P}\{A\} \leq \frac{\mathbb{E}[X_n^+]}{\lambda}.$$

Hence, if $\mathbb{E}[X_n^+]$ is controlled, then we can easily conclude that a larger λ means a lower probability of the process ever crossing it before time n.

The next result follows directly from integrating Theorem 4.2.

Theorem 4.3 ($\mathcal{L}^p$ maximum inequality) *Let $(X_n)_{n\in\mathbb{N}}$ be a submartingale. Then, for $1 < p < \infty$,*

$$\mathbb{E}[(\bar{X}_n)^p] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[(X_n^+)^p].$$

Consequently, if Y_n is a martingale and $Y_n^* = \max_{0\leq m\leq n} |Y_m|$, then

$$\mathbb{E}[|Y_n^*|^p] \leq \left(\frac{p}{p-1}\right)^p \mathbb{E}[|Y_n|^p].$$

Proof. For a proof, see [2], Section 4.4, Theorem 4.4.4. □

This theorem provides an upper bound for the expected size of a process's peak to the power of p. More specifically, it states that if the value of a process at time n is controlled in $\mathcal{L}^p$, then its largest value up to time n is also controlled. However, as we will see in the next example, this inequality is not valid for $p \neq 1$.

Example 4.4 Let us recall Example 3.22. From Theorem 4.1, it follows that, for all n,

$$\mathbb{E}[Y_n] = \mathbb{E}[S_{N\wedge n}] = \mathbb{E}[S_0] = 1.$$

For $M \in \mathbb{N}$, by Theorem 4.8.7 from [2], we have that

$$\mathbb{P}\{\max_{m\geq 0} Y_m \geq M\} = \frac{1}{M}.$$

Therefore,

$$\mathbb{E}\left[\max_{m\geq 0} Y_m\right] = \sum_{M=1}^{\infty} \frac{1}{M} = \infty,$$

and, by the Monotone Convergence Theorem (see [3], Section 2.2, Theorem 2.14), we have that as $n \to \infty$

$$\mathbb{E}\left[\max_{0\leq m\leq n} Y_m\right] \uparrow \infty.$$

Hence, there exists no constant C such that

$$\mathbb{E}\left[\max_{0\leq m\leq n} Y_m\right] \leq C\,\mathbb{E}[Y_n].$$

Theorem 4.3 will be crucial to proving the necessary conditions for convergence in $\mathcal{L}^p$, for $p > 1$, which is stated in the next theorem.

Theorem 4.5 ($\mathcal{L}^p$ Convergence Theorem) *Let $(X_n)_{n\in\mathbb{N}}$ be a martingale with $\mathbb{E}[|X_n|^p] < \infty$, for $p > 1$. Then $X_n \to X$ a.s. and in $\mathcal{L}^p$.*

Proof. For a proof, see [2], Section 4.4, Theorem 4.4.6. □

This theorem shows that $X_n \in \mathcal{L}^p$, for $p > 1$, is a strong enough condition to guarantee not only almost sure convergence of the martingale, but also convergence in $\mathcal{L}^p$.

A natural question is what happens when we only have control in $\mathcal{L}^1$. In this case, boundedness of the expectations alone is not strong enough to guarantee convergence in $\mathcal{L}^1$. To reach a similar convergence theory, one needs a stronger notion, namely uniform integrability. This motivates the next subsection.

4.2. Uniform integrability and convergence in $\mathcal{L}^1$

First, we begin by introducing the concept of uniform integrability, which is one of the key conditions for a martingale to converge in $\mathcal{L}^1$.

Definition 4.6 A collection of random variables $(X_i)_{i\in I}$ is uniformly integrable if

$$\lim_{M\to\infty}\left(\sup_{i\in I}\mathbb{E}[|X_i|\cdot\mathbb{1}_{\{|X_i|\geq M\}}]\right) = 0.$$

Intuitively, uniform integrability ensures that very large values of the random variables do not contribute significantly to the expectation. This is exactly the condition that allows convergence in probability to be upgraded to convergence in $\mathcal{L}^1$, as we will see in the next theorem.

Theorem 4.7 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a sequence of random variables such that $\mathbb{E}[|X_n|] < \infty$ for all n. If $X_n \to X$ in probability, then the following are equivalent:*

(1) $(X_n)_{n\in\mathbb{N}}$ is uniformly integrable;

(2) $X_n \to X$ in $\mathcal{L}^1$;

(3) $\mathbb{E}[|X_n|] \to \mathbb{E}[|X|] < \infty$.

Proof. For a proof, see [2], Section 4.6, Theorem 4.6.2. □

We now have the tools to state and prove the first main result from this section.

Theorem 4.8 *Let* $(X_n)_{n\in\mathbb{N}_0}$ *be a submartingale, then the following are equivalent:*

(1) $(X_n)_{n\in\mathbb{N}_0}$ *is uniformly integrable;*

(2) $(X_n)_{n\in\mathbb{N}_0}$ *converges a.s. and in* $\mathcal{L}^1$*;*

(3) $(X_n)_{n\in\mathbb{N}_0}$ *converges in* $\mathcal{L}^1$*.*

Proof. We start by proving that (1) implies (2). Since $(X_n)_{n\in\mathbb{N}_0}$ is uniformly integrable, then $\mathbb{E}[|X_n|] < \infty$. Hence, by Theorem 3.20, we have that $X_n \to X$ a.s. Note that we are in a probability space. Therefore, by Theorem B.3, we have that $X_n \to X$ in probability, and by Theorem 4.7 it follows that $X_n \to X$ in $\mathcal{L}^1$.

(2) implies (3) trivially.

Finally, to prove that (3) implies (1), note that convergence in $\mathcal{L}^1$ implies convergence in probability (see Theorem B.4). Thus, again from Theorem 4.7 we have that $(X_n)_{n\in\mathbb{N}_0}$ is uniformly integrable. □

This theorem shows that uniform integrability is exactly the missing condition to guarantee convergence in $\mathcal{L}^1$ for submartingales. Now, we want an analogue of the theorem for martingales. For that, we first present the following results, which will be useful later.

Lemma 4.9 *Let* $(\Omega, \mathcal{F}, \mathbb{P})$ *be a probability space,* $X \in \mathcal{L}^1$ *and* $\mathcal{G} \subseteq \mathcal{F}$*. Then the family*

$$\{\mathbb{E}[X \mid \mathcal{G}] : \mathcal{G} \subseteq \mathcal{F} \text{ is a } \sigma\text{-algebra}\}$$

is uniformly integrable.

Proof. For a proof, see [2], Section 4.6, Theorem 4.6.1. □

This result means that, if the random variable is integrable, then its conditional expectations form a uniformly integrable family, even when the conditioning σ-algebra varies.

Lemma 4.10 *If* $(X_n)_{n\in\mathbb{N}}$ *is a martingale such that* $X_n \to X$ *in* $\mathcal{L}^1$*, then* $X_n = \mathbb{E}[X \mid \mathcal{F}_n]$*, for all* n*.*

Proof. For a proof, see [2], Section 4.6, Lemma 4.6.6. □

This lemma shows that a martingale that converges in $\mathcal{L}^1$ is determined by its terminal limit. More precisely, if a martingale converges in $\mathcal{L}^1$ to X, then each X_n can be recovered as the conditional expectation of the limit X given the information available at time n.

Now, we are ready to state and prove the theorem that will give us the necessary conditions for a martingale to converge in $\mathcal{L}^1$.

Theorem 4.11 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a martingale, then the following are equivalent:*

(1) $(X_n)_{n\in\mathbb{N}_0}$ is uniformly integrable;

(2) $(X_n)_{n\in\mathbb{N}_0}$ converges a.s. and in $\mathcal{L}^1$;

(3) $(X_n)_{n\in\mathbb{N}_0}$ converges in $\mathcal{L}^1$;

(4) There is an integrable random variable X such that $X_n = \mathbb{E}[X \mid \mathcal{F}_n]$, for all n.

Proof. We begin by proving that (1) implies (2). Since martingales are submartingales, then we are in the conditions of Theorem 4.8, and it follows directly that $(X_n)_{n\in\mathbb{N}_0}$ converges a.s. and in $\mathcal{L}^1$.

(2) implies (3) trivially.

(3) implies (4) is a direct result of Lemma 4.10.

Finally, (4) implies (1) follows directly from Theorem 4.9. □

Once more, we see that uniform integrability is the key condition for a martingale to converge in $\mathcal{L}^1$. Moreover, we also get that a uniformly integrable martingale can be represented as the conditional expectation of its limiting random variable. In light of this, let us use martingale convergence theorems to study how these conditional expectations behave as the available information increases over time.

Theorem 4.12 *Suppose that $\mathcal{F}_n \uparrow \mathcal{F}_\infty$, i.e. $\mathcal{F}_n$ is an increasing sequence of σ-algebras, and $\mathcal{F}_\infty = \sigma\left(\bigcup_n \mathcal{F}_n\right)$. Then*

$$\mathbb{E}[X \mid \mathcal{F}_n] \xrightarrow[n\to\infty]{} \mathbb{E}[X \mid \mathcal{F}_\infty] \text{ a.s. and in } \mathcal{L}^1.$$

Proof. For a proof, see [2], Section 4.6, Theorem 4.6.8. □

This first result has a very intuitive meaning: as more information becomes available, the conditional expectation approaches the best prediction based on the full information, i.e. $\mathcal{F}_\infty$. A particularly important consequence is obtained when we consider $X = \mathbb{1}_A$.

Theorem 4.13 (Lévy's 0-1 law) *If $\mathcal{F}_n \uparrow \mathcal{F}_\infty$ and $A \in \mathcal{F}_\infty$, then*

$$\mathbb{E}[\mathbb{1}_A \mid \mathcal{F}_n] \to \mathbb{1}_A \text{ a.s.}$$

Proof. For a proof, see [2], Section 4.6, Theorem 4.6.9. □

Intuitively, if $\mathbb{1}_A \in \mathcal{F}_\infty$, then as $\mathcal{F}_n$ increases, our best prediction of $\mathbb{1}_A$ given $\mathcal{F}_n$ becomes increasingly more precise. In the limit, the event A is fully determined, so the conditional expectation should converge to $\mathbb{1}_A$.

Another consequence of Theorem 4.12 is the following.

Theorem 4.14 (Dominated convergence theorem for conditional expectation) *Suppose that $Y_n \to Y$ a.s. and $|Y_n| \leq Z$ for all n where $\mathbb{E}[Z] < \infty$. If $\mathcal{F}_n \to \mathcal{F}_\infty$, then*

$$\mathbb{E}[Y_n \mid \mathcal{F}_n] \to \mathbb{E}[Y \mid \mathcal{F}_\infty] \quad a.s.$$

Proof. For a proof, see [2], Section 4.6, Theorem 4.6.10. □

This theorem guarantees that conditional expectations behave well under limits, provided the variables are controlled by an integrable bound.

5. Optional Stopping Theorems

Having established convergence and integrability results, we can now study one of the central questions in martingale theory: what happens when a martingale is stopped at a random time? The goal of this section will be to study the conditions under which, for a submartingale $(X_n)_{n\in\mathbb{N}_0}$ and stopping times $M \leq N$, the property $\mathbb{E}[X_M] \leq \mathbb{E}[X_N]$ holds. Let us first start with an example.

Example 5.1 Let S_n be the simple random walk from Example 2.1, but with initial position 1, i.e. $S_0 = 1$. Let $N = \inf\{n : S_n = 0\}$. Clearly,

$$\mathbb{E}[S_0] = 1 > 0 = \mathbb{E}[S_N],$$

meaning that it may not always be true that $\mathbb{E}[X_M] \leq \mathbb{E}[X_N]$, for stopping times $M \leq N$. In fact, when N is unbounded, extra conditions are necessary to guarantee the desired property, which will be analysed in this section.

Before considering the unbounded case, let us first look at what happens when N is bounded. Previously, we have already seen with Theorem 4.1 a simple version of a bounded optional stopping result. Let us now generalise this result.

Theorem 5.2 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a submartingale. Let M and N be stopping times such that $M \leq N$ and $\mathbb{P}\{N \leq k\} = 1$. Then*

$$\mathbb{E}[X_M] \leq \mathbb{E}[X_N].$$

Proof. For a proof, see Appendix A.6. □

We have now seen that the desired inequality holds when N is bounded. Next, we present a result that will later be needed to prove the necessary conditions to extend this property to possibly unbounded stopping times.

Theorem 5.3 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a uniformly integrable submartingale. Then for any stopping time N, $X_{N\wedge n}$ is uniformly integrable.*

Proof. For a proof, see [2], Section 4.8, Theorem 4.8.1. □

This theorem shows that, if a submartingale is uniformly integrable, then stopping it at an arbitrary stopping time preserves uniform integrability. This will be used to prove the following theorem, which states that uniform integrability is a sufficient condition for the desired inequality to hold for unbounded stopping times.

Theorem 5.4 *Let $(X_n)_{n\in\mathbb{N}_0}$ be a uniformly integrable submartingale. Then for any stopping time $N \leq \infty$, it holds that $\mathbb{E}[X_0] \leq \mathbb{E}[X_N] \leq \mathbb{E}[X_\infty]$, where $X_\infty = \lim X_n$.*

Proof. From Theorem 4.1, it follows that $\mathbb{E}[X_0] \leq \mathbb{E}[X_{N\wedge n}] \leq \mathbb{E}[X_n]$. Letting $n \to \infty$, from Theorem 4.8, we get that $X_n \to X_\infty$ in $\mathcal{L}^1$. Hence, $\mathbb{E}[X_n] \to \mathbb{E}[X_\infty]$.

Moreover, from Theorem 5.3, we get that $(X_{N\wedge n})_{n\in\mathbb{N}_0}$ is a uniformly integrable submartingale. Therefore, again from Theorem 4.8 it follows that $X_{N\wedge n} \to X_N$ in $\mathcal{L}^1$, thus $\mathbb{E}[X_{N\wedge n}] \to \mathbb{E}[X_N]$.

Hence, as $n \to \infty$, we get $\mathbb{E}[X_0] \leq \mathbb{E}[X_N] \leq \mathbb{E}[X_\infty]$. □

This result states that stopping a submartingale cannot decrease its expected value. In the case of martingales, we have that

$$\mathbb{E}[X_0] = \mathbb{E}[X_N] = \mathbb{E}[X_\infty],$$

meaning that no stopping rule based only on the available information can systematically increase the expected value. In other words, if we have a fair game, then there is no stopping rule that can turn it into a profitable game on average.

6. Branching Processes

We now apply the martingale theory developed so far to a classical model in probability theory: branching processes. These processes are used to model the evolution of a population in which each individual gives birth independently to a random number of descendants.

The simplest and most classical branching process is the Galton–Watson process. We start with one individual at time 0, and this individual has a random number of children. Then, each of those children has a random number of children, independently of each other, with the same distribution, and the procedure continues generation by generation.

Definition 6.1 (Galton–Watson process) Let $\{\xi_i^{(n)} : i, n \geq 1\}$ be a family of independent and identically distributed non-negative integer-valued random variables. The random variable $\xi_i^{(n)}$ represents the number of children produced by the i-th individual in generation $n-1$.

Let

$$p_k = \mathbb{P}\{\xi_i^{(n)} = k\}, \quad k \in \mathbb{N}_0,$$

be the offspring distribution, and assume that $\mu = \mathbb{E}[\xi_i^{(n)}] \in (0, \infty)$. The Galton–Watson process $(Z_n)_{n\in\mathbb{N}_0}$ is defined by $Z_0 = 1$ and

$$Z_{n+1} = \sum_{i=1}^{Z_n} \xi_i^{(n+1)}.$$

If $Z_n = 0$, the sum is empty and we set $Z_{n+1} = 0$.

Here, Z_n represents the number of individuals in generation n and μ is the expected number of children of each individual. Therefore, μ measures the average growth rate of the population. If $\mu < 1$, each individual has on average less than one child, so we expect the population to die out. If $\mu = 1$, the population replaces itself on average. If $\mu > 1$, the population is expected to grow.

Let $\mathcal{F}_n = \sigma\left(\xi_i^{(m)} : 1 \leq m \leq n,\ i \geq 1\right)$. This filtration contains all the information about the reproduction of individuals up to generation n. In particular, Z_n is $\mathcal{F}_n$-measurable.

Proposition 6.2 *For the Galton–Watson process $(Z_n)_{n\in\mathbb{N}_0}$, we have*

$$\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \mu Z_n,$$

and, consequently, $\mathbb{E}[Z_n] = \mu^n$.

Proof. Since Z_n is known given $\mathcal{F}_n$, and since the random variables $\xi_i^{(n+1)}$ are independent of $\mathcal{F}_n$, we get

$$\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \mathbb{E}\left[\sum_{i=1}^{Z_n} \xi_i^{(n+1)} \,\middle|\, \mathcal{F}_n\right] = \sum_{i=1}^{Z_n} \mathbb{E}[\xi_i^{(n+1)}] = \mu Z_n.$$

Taking expectations on both sides and using the tower property it gives

$$\mathbb{E}[Z_{n+1}] = \mu\mathbb{E}[Z_n],$$

and, since $Z_0 = 1$, by induction we get $\mathbb{E}[Z_n] = \mu^n$. □

This result shows that the expected size of the population grows like μ^n. Therefore, instead of studying Z_n directly, it is natural to compare it with its expected order of growth. This motivates the normalised process

$$W_n = \frac{Z_n}{\mu^n}.$$

As we will see, this normalisation produces a martingale.

Lemma 6.3 *The process*

$$W_n := \frac{Z_n}{\mu^n}$$

is a martingale with respect to $(\mathcal{F}_n)_{n\in\mathbb{N}_0}$.

Proof. W_n is integrable and adapted. Furthermore, using Proposition 6.2, we have

$$\mathbb{E}[W_{n+1} \mid \mathcal{F}_n] = \mathbb{E}\left[\frac{Z_{n+1}}{\mu^{n+1}} \,\middle|\, \mathcal{F}_n\right] = \frac{1}{\mu^{n+1}}\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = \frac{\mu Z_n}{\mu^{n+1}} = \frac{Z_n}{\mu^n} = W_n.$$

Therefore, $(W_n)_{n\in\mathbb{N}_0}$ is a martingale. □

Since $W_n \geq 0$, i.e. it is a non-negative martingale, by Corollary 3.21 there exists a random variable W such that $W_n \to W$ a.s. We will now study this limit, beginning with the case where this limit is 0, i.e. when the population gets extinct. Let us first define extinction.

Definition 6.4 (Extinction) The branching process becomes extinct if there exists some generation n such that $Z_n = 0$. Since 0 is an absorbing state, once $Z_n = 0$, we have $Z_m = 0$, for every $m \geq n$. The extinction event is therefore

$$E := \{\exists n \geq 0 : Z_n = 0\}.$$

Let us first look at the case where $\mu < 1$ (subcritical).

Theorem 6.5 *If* $\mu < 1$, *then* $Z_n = 0$ *for all* n *sufficiently large, i.e.* $\mathbb{P}\{E\} = 1$. *Therefore,* $W_n \to 0$.

Proof. Since Z_n is a non-negative integer-valued random variable, we have $\{Z_n > 0\} = \{Z_n \geq 1\}$. Therefore,

$$\mathbb{P}\{Z_n > 0\} = \mathbb{P}\{Z_n \geq 1\} \leq \mathbb{E}[Z_n] = \mu^n,$$

where the last equality comes from Proposition 6.2. If $\mu < 1$, then $\mu^n \to 0$. Hence,

$$\mathbb{P}\{Z_n > 0\} \to 0,$$

or, equivalently,

$$\mathbb{P}\{Z_n = 0\} \to 1.$$

Thus,

$$W_n = \frac{Z_n}{\mu^n} \to 0.$$

□

This result has a very intuitive interpretation. If each individual has fewer than one child on average, then the next generation will always be smaller than the previous one, eventually leading to extinction.

The next result treats a special case when $\mu = 1$ (critical). It is easy to conclude that, if we consider the trivial case where $\mu = 1$ and $\mathbb{P}\{\xi_i^{(n)} = 1\} = 1$, then the population will remain with the same number of individuals forever. Hence, we will see what happens in the critical case if this trivial case is removed, that is, when $\mathbb{P}\{\xi_i^{(n)} = 1\} < 1$.

Theorem 6.6 *Suppose that $\mu = 1$ and $\mathbb{P}\{\xi_i^{(n)} = 1\} < 1$. Then $Z_n = 0$ for all n sufficiently large, i.e. $\mathbb{P}\{E\} = 1$. Therefore, $W_n \to 0$.*

Proof. When $\mu = 1$, the process $(Z_n)_{n\in\mathbb{N}_0}$ itself is a non-negative martingale, because

$$\mathbb{E}[Z_{n+1} \mid \mathcal{F}_n] = Z_n.$$

By Corollary 3.21, there exists a finite random variable Z such that

$$Z_n \to Z \quad \text{a.s.}$$

Since each Z_n is integer-valued, convergence to a finite limit implies that, almost surely, the sequence is eventually constant and equal to its limit, i.e. we must have $Z_n = Z$ for large n.

We now show that the eventual constant cannot be a positive integer. Suppose, by contradiction, that for some $k \geq 1$ the process eventually remains equal to k. If $Z_n = k$, then for Z_{n+1} to also equal k, the total number of children produced by the k individuals in generation n must be exactly k. However, since $\mathbb{P}\{\xi = 1\} < 1$, there exists a natural number $r \neq 1$ such that $\mathbb{P}\{\xi = r\} > 0$. Therefore,

$$\mathbb{P}\{\xi_1 = r, \ldots, \xi_k = r\} = \mathbb{P}\{\xi = r\}^k > 0.$$

On this event, since $r \neq 1$,

$$\xi_1 + \cdots + \xi_k = kr \neq k.$$

Thus,

$$\mathbb{P}\{\xi_1 + \cdots + \xi_k \neq k\} > 0.$$

Therefore,

$$\mathbb{P}\{Z_{n+1} = k \mid Z_n = k\} = \mathbb{P}\{\xi_1 + \cdots + \xi_k = k\} < 1.$$

Consequently, once the process is at level k, there is always a positive probability that it will leave k at the next generation. Since the reproduction variables are independent across generations, the probability that the process remains equal to k forever is 0. Hence, the eventual constant cannot be a positive integer.

Therefore, the only possible limiting value is $Z = 0$, thus $Z_n = 0$ for all sufficiently large n, almost surely. Hence, extinction occurs almost surely and $W_n = Z_n \to 0$. □

We have just concluded that, in both the subcritical case $\mu < 1$ and the non-trivial critical case $\mu = 1$, the population becomes extinct with probability one. The supercritical case $\mu > 1$ is different: there is a positive probability that the population survives forever. To describe this probability, we use generating functions.

Definition 6.7 (Generating function) The generating function of the offspring distribution is the function

$$\varphi(s) = \mathbb{E}[s^\xi] = \sum_{k=0}^{\infty} p_k s^k, \quad 0 \leq s \leq 1.$$

Generating functions are useful because they encode the entire offspring distribution into a single function. In particular, $\varphi(1) = 1$ and

$$\varphi'(1) = \sum_{k=1}^{\infty} k p_k = \mu.$$

Let q be the extinction probability, i.e. $q = \mathbb{P}\{E\}$.

Theorem 6.8 *Suppose that $\mu > 1$. Then q is the unique solution in $[0, 1)$ of $\varphi(\rho) = \rho$.*

Proof. First, note that $\varphi(1) = 1$. For $s < 1$, differentiating term by term gives

$$\varphi'(s) = \sum_{k=1}^{\infty} k p_k s^{k-1} \geq 0,$$

so φ is increasing. Moreover,

$$\lim_{s \uparrow 1} \varphi'(s) = \sum_{k=1}^{\infty} k p_k = \mu.$$

Therefore, as $h \downarrow 0$,

$$\varphi(1) - \varphi(1-h) = \int_{1-h}^{1} \varphi'(s)\, ds \sim \mu h.$$

Since $\mu > 1$, for $h > 0$ sufficiently small we have $\varphi(1-h) < 1 - h$. Also, $\varphi(0) \geq 0$. Hence, by continuity,

there exists at least one solution of $\varphi(x) = x$ in $[0, 1]$.

To prove uniqueness in $[0, 1)$, note that for $s < 1$,

$$\varphi''(s) = \sum_{k=2}^{\infty} k(k-1)p_k s^{k-2} > 0,$$

since $\mu > 1$ implies that $p_k > 0$ for some $k \geq 2$. Thus φ is strictly convex on $[0, 1)$.

Let ρ be the smallest solution of $\varphi(\rho) = \rho$ in $[0, 1)$. Since $\varphi(1) = 1$ and φ is strictly convex, then $\varphi(x) < x$, for $\rho < x < 1$. Therefore, ρ is the only solution of $\varphi(x) = x$ in $[0, 1)$.

We now prove that $\rho = q$. Let

$$q_m := \mathbb{P}(Z_m = 0).$$

We first show that $q_m = \varphi(q_{m-1})$. Indeed, conditioning on Z_1, we obtain

$$q_m = \sum_{k=0}^{\infty} \mathbb{P}\{Z_m = 0 \mid Z_1 = k\}\mathbb{P}\{Z_1 = k\}.$$

If $Z_1 = k$, then extinction by time m occurs if and only if all k families die out in the remaining $m - 1$ generations. By independence, this has probability q_{m-1}^k. Hence

$$q_m = \sum_{k=0}^{\infty} p_k q_{m-1}^k = \varphi(q_{m-1}).$$

Next, we show that $q_m \uparrow \rho$. Clearly, $q_m = \mathbb{P}\{Z_m = 0\}$ is increasing. Moreover, $q_0 = 0 \leq \rho$. If $q_{m-1} \leq \rho$, then, since φ is increasing,

$$q_m = \varphi(q_{m-1}) \leq \varphi(\rho) = \rho.$$

Thus, by induction, $q_m \leq \rho$ for all m. Therefore, since q_m is increasing and bounded above by ρ, the limit

$$q_\infty := \lim_{m\to\infty} q_m$$

exists and satisfies $q_\infty \leq \rho$.

Taking limits in $q_m = \varphi(q_{m-1})$ and using the continuity of φ, we obtain $q_\infty = \varphi(q_\infty)$. Since $q_\infty \leq \rho$ and ρ is the smallest solution of $\varphi(x) = x$ in $[0, 1]$, it follows that $q_\infty = \rho$.

Finally,

$$q = \lim_{m\to\infty} \mathbb{P}\{Z_m = 0\} = \lim_{m\to\infty} q_m = \rho.$$

□

This theorem states that, in the supercritical case, extinction is not certain. Indeed, since q is the unique solution of $\varphi(\rho) = \rho$ in $[0, 1)$, we have $q < 1$. Therefore, $\mathbb{P}\{E^c\} = 1 - q > 0$, i.e. the probability of survival is not zero.

We can summarise the extinction behaviour as follows:

$$\begin{cases} \mu < 1, & \text{extinction occurs almost surely;} \\ \mu = 1 \text{ and } \mathbb{P}\{\xi = 1\} < 1, & \text{extinction occurs almost surely;} \\ \mu > 1, & \text{there is a positive probability of survival.} \end{cases}$$

W_n gives a more refined way to study the supercritical case. Under an additional second moment assumption, we can prove that this martingale is bounded in $\mathcal{L}^2$, and therefore converges not only almost surely but also in $\mathcal{L}^2$.

Theorem 6.9 *Suppose that $\mu > 1$ and that $\mathrm{Var}[\xi_i^{(n)}] = \sigma^2 < \infty$. Then $(W_n)_{n\in\mathbb{N}_0}$ is bounded in $\mathcal{L}^2$. Consequently, there exists a random variable W such that $W_n \to W$ a.s. and in $\mathcal{L}^2$. Moreover, $\mathbb{E}[W] = 1$, so $W \not\equiv 0$.*

Proof. We start by proving that $(W_n)_{n\in\mathbb{N}_0}$ is bounded in $\mathcal{L}^2$.

Since (W_n) is a martingale, we have

$$\mathbb{E}[W_n \mid \mathcal{F}_{n-1}] = W_{n-1}.$$

Therefore, from Theorem B.5, we have

$$\mathbb{E}[W_n^2 \mid \mathcal{F}_{n-1}] = W_{n-1}^2 + \mathbb{E}[(W_n - W_{n-1})^2 \mid \mathcal{F}_{n-1}].$$

Let us first compute the second term. Since

$$W_n - W_{n-1} = \frac{Z_n}{\mu^n} - \frac{Z_{n-1}}{\mu^{n-1}} = \frac{Z_n - \mu Z_{n-1}}{\mu^n},$$

we get

$$\mathbb{E}[(W_n - W_{n-1})^2 \mid \mathcal{F}_{n-1}] = \frac{1}{\mu^{2n}}\mathbb{E}[(Z_n - \mu Z_{n-1})^2 \mid \mathcal{F}_{n-1}].$$

On the event $\{Z_{n-1} = k\}$, we have

$$Z_n = \sum_{i=1}^{k} \xi_i^{(n)}.$$

Hence,

$$Z_n - \mu Z_{n-1} = \sum_{i=1}^{k} (\xi_i^{(n)} - \mu).$$

Since the offspring variables are independent and have variance σ^2,

$$\mathbb{E}[(Z_n - \mu Z_{n-1})^2 \mid \mathcal{F}_{n-1}] = Z_{n-1}\sigma^2.$$

Therefore,

$$\mathbb{E}[(W_n - W_{n-1})^2 \mid \mathcal{F}_{n-1}] = \frac{Z_{n-1}\sigma^2}{\mu^{2n}},$$

and it follows that

$$\mathbb{E}[W_n^2 \mid \mathcal{F}_{n-1}] = W_{n-1}^2 + \frac{Z_{n-1}\sigma^2}{\mu^{2n}}.$$

Taking expectations gives

$$\mathbb{E}[W_n^2] = \mathbb{E}[W_{n-1}^2] + \frac{\sigma^2}{\mu^{2n}}\mathbb{E}[Z_{n-1}] = \mathbb{E}[W_{n-1}^2] + \frac{\sigma^2}{\mu^{n+1}},$$

where the last equality comes from the fact that $\mathbb{E}[Z_{n-1}] = \mu^{n-1}$. Note that $W_0 = 1$, so $\mathbb{E}[W_0^2] = 1$. By induction, we get

$$\mathbb{E}[W_n^2] = 1 + \sigma^2 \sum_{k=2}^{n+1} \mu^{-k}.$$

Since $\mu > 1$, the series

$$\sum_{k=2}^{\infty} \mu^{-k}$$

is convergent. Hence,

$$\sup_n \mathbb{E}[W_n^2] < \infty,$$

meaning that $(W_n)_{n\in\mathbb{N}_0}$ is bounded in $\mathcal{L}^2$. Therefore, by Theorem 4.5 with $p = 2$, there exists a random variable W such that $W_n \to W$ a.s. and in $\mathcal{L}^2$.

Since convergence in $\mathcal{L}^2$ implies convergence in $\mathcal{L}^1$, we also have $W_n \to W$ in $\mathcal{L}^1$. Therefore, $\mathbb{E}[W_n] \to \mathbb{E}[W]$. However, $(W_n)_{n\in\mathbb{N}_0}$ is a martingale and $W_0 = 1$, so $\mathbb{E}[W_n] = \mathbb{E}[W_0] = 1$, for every n. Hence, $\mathbb{E}[W] = 1$. In particular, W cannot be identically equal to 0. □

This theorem strengthens the convergence result obtained from the martingale convergence theorem. We already knew that the martingale converged almost surely, however, the additional finite variance assumption gives convergence in $\mathcal{L}^2$. Consequently, W cannot be identically equal to zero. In other words, in the supercritical case with finite offspring variance, the normalised population size has a non-zero limiting value with positive probability.

Overall, branching processes provide a natural application of martingale theory. In the subcritical case $\mu < 1$, the population becomes extinct almost surely because its expected size tends to zero. In the non-trivial critical case $\mu = 1$, martingale convergence shows that the population also becomes extinct almost surely. In the supercritical case $\mu > 1$, generating functions show that extinction no longer occurs with probability one. Instead, there is a positive probability of survival. Finally, the martingale $W_n = Z_n/\mu^n$ allows us to study the growth of the population on the survival event, and under a finite variance condition this normalised population converges to a non-trivial limit.

7. Conclusion

In this text, we introduced the theory of discrete-time martingales, starting from the notion of conditional expectation, which allowed us to formalise the idea of updating predictions as new information becomes available. From there, we defined martingales, submartingales and supermartingales, interpreting them respectively as fair, favourable and unfavourable processes.

We then studied several structural properties of martingales. In particular, we saw that martingales are stable under stopping and that convex transformations naturally lead to submartingales. These ideas culminated in the upcrossing inequality, which controls the number of oscillations of a submartingale between two fixed levels. This was the key tool used to prove the martingale convergence theorem, showing that suitable boundedness assumptions force almost sure convergence.

After establishing almost sure convergence, we investigated stronger modes of convergence. Doob's inequality allowed us to control the maximum of a martingale in $\mathcal{L}^p$, leading to the $\mathcal{L}^p$ convergence theorem for $p > 1$. We then introduced uniform integrability, which plays the corresponding role in $\mathcal{L}^1$. In particular, we saw that uniform integrability is precisely the additional condition needed to upgrade almost sure convergence to convergence in $\mathcal{L}^1$ for submartingales.

We also studied optional stopping theorems, which address one of the central questions in martingale theory: under what conditions can a martingale be stopped at a random time while preserving its properties? The examples showed that such results cannot hold without additional assumptions, namely for unbounded stopping times. However, under boundedness or uniform integrability conditions, the expected value inequalities associated with submartingales are preserved.

Finally, we applied the theory to branching processes. The Galton–Watson process provided a concrete example where martingales arise naturally through the normalised population size $W_n = Z_n/\mu^n$. Using martingale convergence results, we were able to study the long-term behaviour of the population. In the subcritical and non-trivial critical cases, extinction occurs almost surely, while in the supercritical case there is a positive probability of survival. Under additional moment assumptions, the martingale W_n converges to a non-trivial limit, giving a more refined description of the population growth.

Overall, martingales provide a powerful framework for studying stochastic processes over time. They connect conditional expectation, convergence and stopping times in a unified way. Although the theory originates from simple gambling ideas, the results developed here show how martingales become a flexible and rigorous tool for analysing the long-term behaviour of random systems.

A. Additional Proofs

A.1 Proof of Proposition 2.4

Let us start by proving uniqueness. Let Y and Y' be two versions of $\mathbb{E}[X \mid \mathcal{F}]$. Consider the set $A = \{Y - Y' > 0\}$. By the definition of conditional expectation, we have

$$\int_A Y \, \mathrm{d}\mathbb{P} = \int_A Y' \, \mathrm{d}\mathbb{P} \Leftrightarrow \int_A Y - Y' \, \mathrm{d}\mathbb{P} = 0.$$

However, the integral of a strictly positive measurable function over a set with positive measure is strictly positive. Therefore, this is only possible if $\mathbb{P}\{A\} = 0$, so $Y \leq Y'$ a.s. Now, we repeat this procedure with the set $B = \{Y' - Y > 0\}$ and we obtain $Y \geq Y'$ a.s. Hence, $Y = Y'$ a.s.

Now, let us prove existence. Suppose that $X \geq 0$ is an integrable random variable. Define a measure ν on $\mathcal{F}$ by

$$\nu(A) := \int_A X \, \mathrm{d}\mathbb{P}, \quad A \in \mathcal{F}.$$

Since $\nu \ll \mathbb{P}$ (see Appendix B.1), by the Radon-Nikodym theorem (see [3], Section 3.2, Theorem 3.8.) there exists a measurable function $Y = \mathrm{d}\nu/\mathrm{d}\mathbb{P}$ (Radon-Nikodym derivative) such that

$$\nu(A) = \int_A Y \, \mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{F}.$$

By construction, $Y \geq 0$ and satisfies

$$\int_A X \, \mathrm{d}\mathbb{P} = \int_A Y \, \mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{F},$$

so Y is a version of $\mathbb{E}[X \mid \mathcal{F}]$.

For the general case, write $X = X^+ - X^-$, where $X^+ = (X)^+$ and $X^- = (-X)^+$, and apply the same construction to X^+ and X^-. Let Y_1 and Y_2 be the Radon-Nikodym derivatives for X^+ and X^-. Then $Y_1 - Y_2 \in \mathcal{F}$ is integrable and for all $A \in \mathcal{F}$ we have

$$\int_A X \, \mathrm{d}\mathbb{P} = \int_A X^+ \, \mathrm{d}\mathbb{P} - \int_A X^- \, \mathrm{d}\mathbb{P} = \int_A Y_1 \, \mathrm{d}\mathbb{P} - \int_A Y_2 \, \mathrm{d}\mathbb{P} = \int_A (Y_1 - Y_2) \, \mathrm{d}\mathbb{P}.$$

Hence, $Y_1 - Y_2$ is a version of $\mathbb{E}[X \mid \mathcal{F}]$. □

A.2 Proof of Theorem 2.5 (5 to 8)

- **Proof of property 5.**

  Let us start by proving (i). Let X be an integrable random variable. Then, by Lemma 2.3, $\mathbb{E}[X \mid \mathcal{F}]$ is an integrable random variable and, by the definition of conditional expectation, we have

  $$\int_A X \, \mathrm{d}\mathbb{P} = \int_A \mathbb{E}[X \mid \mathcal{F}] \, \mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{F}$$

and

$$\int_B \mathbb{E}[X \mid \mathcal{F}]\,\mathrm{d}\mathbb{P} = \int_B \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}] \mid \mathcal{G}]\,\mathrm{d}\mathbb{P}, \quad \forall B \in \mathcal{G}.$$

Given that $A \in \mathcal{F} \subseteq \mathcal{G}$, it follows that

$$\int_A X\,\mathrm{d}\mathbb{P} = \int_A \mathbb{E}[X \mid \mathcal{F}]\,\mathrm{d}\mathbb{P} = \int_A \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}] \mid \mathcal{G}]\,\mathrm{d}\mathbb{P},$$

hence $\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}] \mid \mathcal{G}]$ is a version of the conditional expectation of X given $\mathcal{F}$, i.e.

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{F}] \mid \mathcal{G}] = \mathbb{E}[X \mid \mathcal{F}].$$

To prove (ii), we follow a similar rationale. By the definition of conditional expectation, we have

$$\int_A \mathbb{E}[X \mid \mathcal{G}]\,\mathrm{d}\mathbb{P} = \int_A \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{F}]\,\mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{F}.$$

Since $A \in \mathcal{F} \subseteq \mathcal{G}$, the following holds true

$$\int_A X\,\mathrm{d}\mathbb{P} = \int_A \mathbb{E}[X \mid \mathcal{G}]\,\mathrm{d}\mathbb{P} = \int_A \mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{F}]\,\mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{F}.$$

Hence,

$$\mathbb{E}[\mathbb{E}[X \mid \mathcal{G}] \mid \mathcal{F}] = \mathbb{E}[X \mid \mathcal{F}].$$

□

- **Proof of property 6.**

  Let X_1, X_2 be integrable random variables, $\mathcal{F}$ a σ-algebra and $Z = \mathbb{E}[X_2 \mid \mathcal{F}]$. We want to show that $X_1 Z$ is the conditional expectation of $X_1 X_2$ given $\mathcal{F}$, i.e.

  $$\int_A X_1 Z\,\mathrm{d}\mathbb{P} = \int_A X_1 X_2\,\mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{F}$$

  First, let us prove that the equality holds when X_1 is an indicator function. Let $X_1 = \mathbb{1}_B$ for some $B \in \mathcal{F}$. Then

  $$\int_A X_1 Z\,\mathrm{d}\mathbb{P} = \int_A \mathbb{1}_B Z\,\mathrm{d}\mathbb{P} = \int_{A \cap B} Z\,\mathrm{d}\mathbb{P}.$$

  Since $A \cap B \in \mathcal{F}$, we know that

  $$\int_{A \cap B} Z\,\mathrm{d}\mathbb{P} = \int_{A \cap B} X_2\,\mathrm{d}\mathbb{P}$$

  and thus

  $$\int_A X_1 Z\,\mathrm{d}\mathbb{P} = \int_{A \cap B} X_2\,\mathrm{d}\mathbb{P} = \int_A \mathbb{1}_B X_2\,\mathrm{d}\mathbb{P} = \int_A X_1 X_2\,\mathrm{d}\mathbb{P}, \quad \forall A \in \mathcal{F}.$$

  We next prove the statement for non-negative simple functions. Let $X_1 = \sum_{i=1}^n c_i \mathbb{1}_{B_i}$, with $c_i \geq 0$

and $B_i \in \mathcal{F}$. Then

$$\int_A X_1 Z \, \mathrm{d}\mathbb{P} = \int_A \sum_{i=1}^{n} c_i \mathbb{1}_{B_i} Z \, \mathrm{d}\mathbb{P} = \sum_{i=1}^{n} c_i \int_A \mathbb{1}_{B_i} Z \, \mathrm{d}\mathbb{P}.$$

From the argument above, it follows that

$$\sum_{i=1}^{n} c_i \int_A \mathbb{1}_{B_i} Z \, \mathrm{d}\mathbb{P} = \sum_{i=1}^{n} c_i \int_A \mathbb{1}_{B_i} X_2 \, \mathrm{d}\mathbb{P} = \int_A \sum_{i=1}^{n} c_i \mathbb{1}_{B_i} X_2 \, \mathrm{d}\mathbb{P} = \int_A X_1 X_2 \, \mathrm{d}\mathbb{P}.$$

We now extend the result for when X_1 is any non-negative function. By the Approximation Theorem (see [3], Section 2.1, Theorem 2.10), there exists a sequence of non-negative simple functions $\{X_{1n}\}_{n\geq 1}$ such that $X_{1n} \uparrow X_1$. Thus

$$\int_A X_1 Z \, \mathrm{d}\mathbb{P} = \int_A \lim_n X_{1n} \, Z \, \mathrm{d}\mathbb{P}.$$

By the Monotone Convergence Theorem (see [3], Section 2.2, Theorem 2.14) and, applying the result obtained above, we have

$$\int_A \lim_n X_{1n} \, Z \, \mathrm{d}\mathbb{P} = \lim_n \int_A X_{1n} \, Z \, \mathrm{d}\mathbb{P} = \lim_n \int_A X_{1n} \, X_2 \, \mathrm{d}\mathbb{P} = \int_A \lim_n X_{1n} \, X_2 \, \mathrm{d}\mathbb{P} = \int_A X_1 \, X_2 \, \mathrm{d}\mathbb{P}.$$

Lastly, to generalise for any integrable random variable X_1, we use the decomposition $X_1 = X_1^+ - X^-$, where $X_1^+ = (X_1)^+$ and $X_1^- = (-X_1)^+$, and by the previous point it follows that

$$\int_A X_1 Z \, \mathrm{d}\mathbb{P} = \int_A X_1^+ Z \, \mathrm{d}\mathbb{P} - \int_A X_1^- Z \, \mathrm{d}\mathbb{P} = \int_A X_1^+ X_2 \, \mathrm{d}\mathbb{P} - \int_A X_1^- X_2 \, \mathrm{d}\mathbb{P} = \int_A X_1 X_2 \, \mathrm{d}\mathbb{P}.$$

Hence, $X_1 Z$ is the conditional expectation of $X_1 X_2$, i.e. $\mathbb{E}[X_1 X_2 \mid \mathcal{F}] = X_1 \mathbb{E}[X_2 \mid \mathcal{F}]$.

□

- **Proof of property 7.**

  The result follows from the same argument as the usual Cauchy–Schwarz inequality. □

- **Proof of property 8.**

  Let X be an integrable random variable and independent from a σ-algebra $\mathcal{F}$. We want to show that

  $$\mathbb{E}[X \mid \mathcal{F}] = \mathbb{E}[X],$$

  that is, $\mathbb{E}[X]$ is both $\mathcal{F}$-measurable and that for all $A \in \mathcal{F}$ we have

  $$\int_A \mathbb{E}[X] \mathrm{d}\mathbb{P} = \int_A X \mathrm{d}\mathbb{P}.$$

  The first point is immediate, as $\mathbb{E}[X]$ is a constant, thus $\mathbb{E}[X]$ is $\mathcal{F}$-measurable. To prove the second

point, let us use again the fact that $\mathbb{E}[X]$ is a constant and thus

$$\int_A \mathbb{E}[X]\mathrm{d}\mathbb{P} = \mathbb{E}[X]\int_\Omega \mathbb{1}_A \mathrm{d}\mathbb{P} = \mathbb{E}[X]\cdot\mathbb{P}\{A\}.$$

Now, we just need to show that

$$\int_A X \ \mathrm{d}\mathbb{P} = \mathbb{E}[X]\cdot\mathbb{P}\{A\}.$$

We start by writing

$$\int_A X \ \mathrm{d}\mathbb{P} = \int_\Omega X\cdot\mathbb{1}_A \ \mathrm{d}\mathbb{P},$$

and since X is independent from $\mathcal{F}$, then X is independent from any $A\in\mathcal{F}$. Therefore,

$$\int_\Omega X\cdot\mathbb{1}_A \ \mathrm{d}\mathbb{P} = \left(\int_\Omega X \ \mathrm{d}\mathbb{P}\right)\cdot\left(\int_\Omega \mathbb{1}_A \ \mathrm{d}\mathbb{P}\right) = \mathbb{E}[X]\cdot\mathbb{P}\{A\},$$

which ends the proof. □

A.3 Proof of Proposition 3.3

Since $(X_n)_{n\in\mathbb{N}_0}$ is a martingale, we have

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$$

for all n. Taking expectations on both sides and using the tower property, we get

$$\mathbb{E}[X_{n+1}] = \mathbb{E}\left[\mathbb{E}[X_{n+1} \mid \mathcal{F}_n]\right] = \mathbb{E}[X_n].$$

Finally, we conclude by induction that

$$\mathbb{E}[X_n] = \mathbb{E}[X_0]$$

for all $n\in\mathbb{N}_0$. □

A.4 Proof of Corollary 3.11

Let $(X_n)_{n\in\mathbb{N}_0}$ be a supermartingale and φ an increasing concave function with $\mathbb{E}[|\varphi(X_n)|]<\infty$ for all n. Then, by Jensen's inequality (see Appendix B.2), it follows that

$$\varphi(X_n) \geq \varphi(\mathbb{E}[X_{n+1} \mid \mathcal{F}_n]) \geq \mathbb{E}[\varphi(X_{n+1}) \mid \mathcal{F}_n].$$

Hence, $(\varphi(X_n))_{n\in\mathbb{N}_0}$ is a supermartingale. □

A.5 Proof of Theorem 3.15

We will prove this theorem for martingales. The proof for submartingales and supermartingales is anal-

ogous to this one, replacing = with $\geq$ and $\leq$, respectively. Let $(X_n)_{n\in\mathbb{N}_0}$ be a martingale. Then

$$|X_{N\wedge n}| \leq \max_{0\leq i\leq n} |X_i| \leq \sum_{i=0}^{n} |X_i|$$

and since all X_i are integrable we have

$$\int_\Omega |X_{N\wedge n}| \, \mathrm{d}\mathbb{P} \leq \int_\Omega \sum_{i=0}^{n} |X_i| \, \mathrm{d}\mathbb{P} \leq \sum_{i=0}^{n} \int_\Omega |X_i| \, \mathrm{d}\mathbb{P} < \infty.$$

Moreover, we can write $X_{N\wedge n}$ as

$$X_{N\wedge n} = \sum_{i=0}^{n-1} X_i \mathbb{1}_{\{N=i\}} + X_n \mathbb{1}_{\{N\geq n\}}.$$

Note that $\mathbb{1}_{\{N=i\}} \in \mathcal{F}_n$ and $\mathbb{1}_{\{N\geq n\}} \in \mathcal{F}_n$. Therefore, we can conclude that $X_{N\wedge n}$ is $\mathcal{F}_n$-measurable, thus $(X_{N\wedge n})_{n\in\mathbb{N}_0}$ is an adapted integrable process. Now, let us prove the martingale property.

$$\mathbb{E}[X_{N\wedge(n+1)} \mid \mathcal{F}_n] = \mathbb{E}\left[\sum_{i=0}^{n} X_i \mathbb{1}_{\{N=i\}} + X_{n+1}\mathbb{1}_{\{N\geq n+1\}} \Big| \mathcal{F}_n\right] = \sum_{i=0}^{n} \mathbb{E}\left[X_i \mathbb{1}_{\{N=i\}} \mid \mathcal{F}_n\right] + \mathbb{E}[X_{n+1}\mathbb{1}_{\{N\geq n+1\}} \mid \mathcal{F}_n].$$

We know that $X_i \mathbb{1}_{\{N=i\}} \in \mathcal{F}_n$ for $i \leq n$, so the first term becomes

$$\sum_{i=0}^{n} \mathbb{E}\left[X_i \mathbb{1}_{\{N=i\}} \mid \mathcal{F}_n\right] = \sum_{i=0}^{n} X_i \mathbb{1}_{\{N=i\}} = \sum_{i=0}^{n-1} X_i \mathbb{1}_{\{N=i\}} + X_n \mathbb{1}_{\{N=n\}}.$$

In the second term, let us rewrite $\{N \geq n+1\} = \{N > n\}$, which is clearly $\mathcal{F}_n$-measurable. Hence

$$\mathbb{E}[X_{n+1}\mathbb{1}_{\{N>n\}} \mid \mathcal{F}_n] = \mathbb{1}_{\{N>n\}} \cdot \mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = \mathbb{1}_{\{N>n\}} X_n.$$

Putting all together, we have

$$\mathbb{E}[X_{N\wedge(n+1)} \mid \mathcal{F}_n] = \sum_{i=0}^{n-1} X_i \mathbb{1}_{\{N=i\}} + X_n \mathbb{1}_{\{N=n\}} + X_n \mathbb{1}_{\{N>n\}} = \sum_{i=0}^{n-1} X_i \mathbb{1}_{\{N=i\}} + X_n \mathbb{1}_{\{N\geq n\}} = X_{N\wedge n}.$$

□

A.6 Proof of Theorem 5.2

Since $M \leq N$ and $\mathbb{P}\{N \leq k\} = 1$, both stopping times are bounded by k. Define,

$$H_i = \mathbf{1}_{\{M<i\leq N\}}, \quad \text{for } i = 1, \ldots, k.$$

We first note that $(H_i)_{i\leq k}$ is predictable. Indeed, $\{M < i \leq N\} = \{M \leq i-1\} \cap \{N \geq i\}$, and since M and N are stopping times, we have $\{M \leq i-1\} \in \mathcal{F}_{i-1}$ and $\{N \geq i\} = \{N \leq i-1\}^c \in \mathcal{F}_{i-1}$. Therefore, $\{M < i \leq N\} \in \mathcal{F}_{i-1}$, so H_i is $\mathcal{F}_{i-1}$-measurable.

Now observe that

$$X_N - X_M = \sum_{i=1}^{k} H_i(X_i - X_{i-1}).$$

Indeed, for each $\omega \in \Omega$, the sum only includes the increments from $M(\omega)+1$ up to $N(\omega)$ and therefore

$$\sum_{i=M(\omega)+1}^{N(\omega)} (X_i(\omega) - X_{i-1}(\omega)) = X_{N(\omega)}(\omega) - X_{M(\omega)}(\omega).$$

Taking expectations, we obtain

$$\mathbb{E}[X_N - X_M] = \sum_{i=1}^{k} \mathbb{E}\left[H_i(X_i - X_{i-1})\right].$$

Since H_i is $\mathcal{F}_{i-1}$-measurable and bounded, we have

$$\mathbb{E}\left[H_i(X_i - X_{i-1})\right] = \mathbb{E}\left[H_i\, \mathbb{E}[X_i - X_{i-1} \mid \mathcal{F}_{i-1}]\right].$$

Since $(X_n)_{n\in\mathbb{N}_0}$ is a submartingale we have

$$\mathbb{E}[X_i - X_{i-1} \mid \mathcal{F}_{i-1}] \geq 0,$$

and, together with the fact that $H_i \geq 0$, it follows that

$$\mathbb{E}\left[H_i(X_i - X_{i-1})\right] \geq 0.$$

Therefore,

$$\mathbb{E}[X_N - X_M] \geq 0 \Leftrightarrow \mathbb{E}[X_M] \leq \mathbb{E}[X_N].$$

□

B. Additional Theorems

Theorem B.1 *Let μ and ν be measures, and ν defined by*

$$\nu(A) := \int_A f \, \mathrm{d}\mu, \quad A \in \mathcal{F}.$$

Then $\nu \ll \mu$.

Proof. Let $A \in \mathcal{F}$ and $f \in \mathcal{F}$ a μ-integrable function. We want to show that if $\mu(A) = 0$, then $\nu(A) = 0$, i.e.

$$\int_A f \, \mathrm{d}\mu = 0$$

Let us first consider f as an indicator function of any set $E \in \mathcal{F}$, i.e. $f = \mathbb{1}_E$. Then

$$\int_A f \, \mathrm{d}\mu = \int_A \mathbb{1}_E \, \mathrm{d}\mu = \int_{A \cap E} 1 \, \mathrm{d}\mu \leq \mu(A) = 0$$

Given that $\nu(A) \geq 0$, we obtain $\nu(A) = 0$.

We can extend the result to non-negative simple functions. Let $f = \sum_{i=1}^n c_i \mathbb{1}_{E_i}$, with $c_i \geq 0$ and $E_i \in \mathcal{F}$. Then

$$\int_A f \, \mathrm{d}\mu = \int_A \sum_{i=1}^n c_i \mathbb{1}_{E_i} \, \mathrm{d}\mu = \sum_{i=1}^n c_i \int_A \mathbb{1}_{E_i} \, \mathrm{d}\mu.$$

From the result above, it follows that

$$\int_A f \, \mathrm{d}\mu = \sum_{i=1}^n c_i \cdot 0 = 0.$$

We next consider the case where f is any non-negative μ-integrable function. By the Approximation Theorem (see [3], Section 2.1, Theorem 2.10), there exists a sequence of non-negative simple functions $\{f_n\}_{n \geq 1}$ such that $f_n \uparrow f$. Therefore,

$$\int_A f \, \mathrm{d}\mu = \int_A \lim_n f_n \, \mathrm{d}\mu$$

and using the Monotone Convergence Theorem (see [3], Section 2.2, Theorem 2.14) and the previous point, we have

$$\int_A \lim_n f_n \, \mathrm{d}\mu = \lim_n \int_A f_n \, \mathrm{d}\mu = 0.$$

Lastly, let us generalise the result to any μ-integrable function f. We can consider the following decomposition $f = f^+ - f^-$, where $f^+ = (f)^+$ and $f^- = (-f)^+$, and it follows from the previous step that

$$\int_A f \, \mathrm{d}\mu = \int_A f^+ \, \mathrm{d}\mu - \int_A f^- \, \mathrm{d}\mu = 0.$$

Hence, $\nu \ll \mu$. □

Theorem B.2 (Jensen's Inequality) *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability measure, $f \in \mathcal{L}^1(\mathbb{P})$, and let $\varphi : \mathbb{R} \to \mathbb{R}$ be a convex function. Then*

$$\varphi\left(\int_\Omega f \, \mathrm{d}\mathbb{P}\right) \leq \int_\Omega \varphi(f) \, \mathrm{d}\mathbb{P}.$$

If φ is concave, the inequality is reversed:

$$\varphi\left(\int_\Omega f \, \mathrm{d}\mathbb{P}\right) \geq \int_\Omega \varphi(f) \, \mathrm{d}\mathbb{P}.$$

Proof. Let us start with the case for convex functions. If φ is a convex function, then at each $x_0 \in \mathbb{R}$ there exists $a, b \in \mathbb{R}$ such that $\varphi(x_0) = ax_0 + b$ and $\varphi(x) \geq ax + b$, for all $x \in \mathbb{R}$. Let $x_0 = \int_\Omega f \, \mathrm{d}\mathbb{P}$. Then,

$$\varphi\left(\int_\Omega f \, \mathrm{d}\mathbb{P}\right) = a \int_\Omega f \, \mathrm{d}\mathbb{P} + b.$$

Since $\mathbb{P}(\Omega) = 1$, then

$$a \int_\Omega f \, \mathrm{d}\mathbb{P} + b = \int_\Omega af \, \mathrm{d}\mathbb{P} + \int_\Omega b \, \mathrm{d}\mathbb{P} = \int_\Omega af + b \, \mathrm{d}\mathbb{P} \leq \int_\Omega \varphi(f) \, \mathrm{d}\mathbb{P}.$$

Now, let φ be concave, then $-\varphi$ is convex. Hence

$$-\varphi\left(\int_\Omega f \, \mathrm{d}\mathbb{P}\right) \leq \int_\Omega -\varphi(f) \, \mathrm{d}\mathbb{P} \Leftrightarrow \varphi\left(\int_\Omega f \, \mathrm{d}\mathbb{P}\right) \geq \int_\Omega \varphi(f) \, \mathrm{d}\mathbb{P}.$$

□

Theorem B.3 *Let $(\Omega, \mathcal{F}, \mu)$ be a probability space, and let $\{f_n\}_n$ be a sequence of measurable functions such that $f_n \to f$ almost surely. Then, $f_n \to f$ in probability.*

Proof. By Egoroff's theorem (see [3], Section 2.4, Theorem 2.33), since $\mu(\Omega) < \infty$ and $f_n \to f$ almost everywhere, we know that $f_n \to f$ almost uniformly. We want to prove that $f_n \to f$ in probability. That is, for every $\varepsilon > 0$, as $n \to \infty$ we have

$$\mu\left(\{x \in \Omega : |f_n(x) - f(x)| > \varepsilon\}\right) \to 0.$$

Let $\varepsilon > 0$ and $\delta > 0$. By almost uniform convergence, there exists $E_\delta \in \mathcal{F}$ such that $\mu(E_\delta^c) \leq \delta$ and $f_n \to f$ uniformly on E_δ. Hence, there exists $N = N(\varepsilon, \delta) \in \mathbb{N}$ such that, for every $n \geq N$,

$$\sup_{x \in E_\delta} |f_n(x) - f(x)| \leq \varepsilon.$$

Therefore, for every $n \geq N$, if

$$|f_n(x) - f(x)| > \varepsilon,$$

then $x \notin E_\delta$. Hence,

$$\{x \in \Omega : |f_n(x) - f(x)| > \varepsilon\} \subseteq E_\delta^c.$$

It follows that, for every $n \geq N$,

$$\mu\left(\{x \in \Omega : |f_n(x) - f(x)| > \varepsilon\}\right) \leq \mu(E_\delta^c) \leq \delta.$$

Since $\delta > 0$ was arbitrary, we conclude that

$$\mu\left(\{x \in \Omega : |f_n(x) - f(x)| > \varepsilon\}\right) \to 0.$$

Therefore, $f_n \to f$ in probability. □

Theorem B.4 *Let $\{f_n\}_n$ be a sequence of measurable functions such that $f_n \to f$ in $\mathcal{L}^p$. Then, $f_n \to f$ in measure.*

Proof. We want to prove that, for all $\varepsilon > 0$,

$$\lim_{n\to\infty} \mu(\{x \in \Omega : |f_n(x) - f(x)| > \varepsilon\}) = 0.$$

Let $A_n^\varepsilon = \{x \in \Omega : |f_n(x) - f(x)| > \varepsilon\}$. Then, in A_n^ε we have

$$|f_n(x) - f(x)| > \varepsilon \Leftrightarrow \frac{1}{\varepsilon^p}|f_n(x) - f(x)|^p > 1.$$

Note that we can write

$$\mathbb{1}_{A_n^\varepsilon} < \frac{1}{\varepsilon^p}|f_n(x) - f(x)|^p,$$

since $\mathbb{1}_{A_n^\varepsilon} = 1$ when $x \in A_n^\varepsilon$ and 0 otherwise. Taking integrals on both sides we get

$$\mu(A_n^\varepsilon) < \frac{1}{\varepsilon^p}\int_\Omega |f_n(x) - f(x)|^p \, \mathrm{d}\mu \to 0,$$

when $n \to \infty$, since $f_n \to f$ in $\mathcal{L}^p$, which ends the proof.

□

Theorem B.5 *If $(X_n)_{n\in\mathbb{N}_0}$ is a martingale such that $\mathbb{E}[X_n^2] < \infty$ for all n, then*

$$\mathbb{E}[(X_n - X_m)^2 \mid \mathcal{F}_m] = \mathbb{E}[X_n^2 \mid \mathcal{F}_m] - X_m^2.$$

Proof. Using the linearity of conditional expectation, we have

$$\mathbb{E}[(X_n - X_m)^2 \mid \mathcal{F}_m] = \mathbb{E}[X_n^2 - 2X_nX_m + X_m^2 \mid \mathcal{F}_m].$$

Hence,

$$\mathbb{E}[(X_n - X_m)^2 \mid \mathcal{F}_m] = \mathbb{E}[X_n^2 \mid \mathcal{F}_m] - 2\mathbb{E}[X_nX_m \mid \mathcal{F}_m] + \mathbb{E}[X_m^2 \mid \mathcal{F}_m].$$

From property 6 from Theorem 2.5, we have

$$\mathbb{E}[X_nX_m \mid \mathcal{F}_m] = X_m\mathbb{E}[X_n \mid \mathcal{F}_m],$$

and

$$\mathbb{E}[X_m^2 \mid \mathcal{F}_m] = X_m^2.$$

Moreover, since (X_n) is a martingale and $m \leq n$,

$$\mathbb{E}[X_n \mid \mathcal{F}_m] = X_m.$$

Putting these identities together, we have

$$\mathbb{E}[(X_n - X_m)^2 \mid \mathcal{F}_m] = \mathbb{E}[X_n^2 \mid \mathcal{F}_m] - 2X_m^2 + X_m^2.$$

Thus,

$$\mathbb{E}[(X_n - X_m)^2 \mid \mathcal{F}_m] = \mathbb{E}[X_n^2 \mid \mathcal{F}_m] - X_m^2.$$

□

References

[1] J. L. Doob. What is a martingale? *The American Mathematical Monthly*, 78(5):451–463, 1971.

[2] Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, 5th edition, 2019.

[3] Gerald B. Folland. *Real Analysis: Modern Techniques and Their Applications*. John Wiley & Sons, Inc., New York, 2nd edition, 1999.